Chapter 33

Campbell Biology, AP Edition, 12e

Photo of blue dragon (Glaucus atlanticus) floating in water shows a bright blue body that tapers to a point, with spiderlike appendages.

Chapter 33

An Introduction to Invertebrates

Figure 33.1

The blue dragon (*Glaucus atlanticus*) has thin, finger-like structures that help it to float (upside down) on the sea's surface. It eats Portuguese men-of-war and absorbs their stinging cells, which it then uses to deliver a deadly sting of its own. The blue dragon is just one of more than a million species of invertebrates (animals without a backbone), which account for over 95% of animal species.

Tree diagram showing animal phylogeny starting with ancestral protist.

Tree diagram for animal phylogeny represented as a list.

Ancestral Protist = Common ancestor of all animals (Branch Point 1). Porifera (sponge drawn).

(Eumetazoa) Branch point 2. Cnidaria (Hydra drawn).

(Bilateria) Branch point 3

Branch point 4. Lophotrochozoa (snail drawn). Ecdysozoa (crayfish drawn). Deuterostomia (fish drawn).

Key Concepts

- **33.1Sponges are basal animals that lack tissues****
- **33.2Cnidarians are an ancient phylum of eumetazoans****

- **33.3Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms****
- **33.4Ecdysozoans are the most species-rich animal group****
- **33.5Echinoderms and chordates are deuterostomes****

Advanced placement edition

Through natural selection, invertebrates have adapted to almost every habitat on Earth **(BIG IDEA 1)**. Invertebrates account for the majority of known animal species, and they interact with virtually all organisms on Earth **(BIG IDEA 4)**.

Study Tip

Make a table: Invertebrates differ in how they obtain food, remove wastes, reproduce, and move throughout their habitat. As you read, summarize how members of the 11 phyla of invertebrates described in detail in this chapter solve these shared challenges of life.

A table lists phylum, feeding method, waste removal, reproduction, and movement.

The table is as follows:

Phylum	Feeding method	Waste removal	Reproduction	Movement
Porifera	Filter feeders	Diffusion	Hermaphrodites	Sessile
blank	blank	blank	blank	blank

****Invertebrates** are animals that lack a backbone. They occupy almost every habitat on Earth, from the scalding water released by deep-sea hydrothermal vents to the frozen ground of Antarctica. Evolution in these varied environments has produced an immense diversity of forms, including species that are microscopic and species whose members can grow to 18 m long (1.5 times the length of a school bus). In this chapter, we'll take a tour of the invertebrates, using the phylogenetic tree in Figure 33.1 as a guide. **Figure 33.2** surveys 23 phyla as representatives of invertebrate diversity.

Figure 33.2

Exploring Invertebrate Diversity

Exploring invertebrate and vertebrate diversity.

Kingdom Animalia encompasses 1.3 million known species, and estimates of the total number of species range as high as 10–20 million. Of the 23 phyla surveyed here, 13 are discussed more fully in this chapter, Chapter 32, or Chapter 34; cross-references are given at the end of their descriptions.

Porifera (5,500 species): Animals in this phylum are informally called sponges. Sponges are sessile animals that lack tissues. They live as filter feeders, trapping particles that pass through the internal channels of their body (see Concept 33.1). A photograph of a sponge is shown.

Placozoa (1 species): The single known species in this phylum, Trichoplax adhaerens, doesn't look like an animal. It consists of a simple bilayer of a few thousand cells and reproduces by dividing into two individuals or by budding off many multicellular individuals. Placozoans are thought to be basal animals, but it is not yet known how they are related to Porifera, Cnidaria, and other early-diverging phyla. A light microscope image of a placozoan is shown and a scale bar reads 0.5 m m.

Cnidaria (10,000 species): Cnidarians include corals, jellies, and hydras. These animals have a diploblastic, radially symmetrical body plan that includes a gastrovascular cavity with a single opening that serves as both mouth and anus (see Concept 33.2). A photograph of a jelly is shown.

Acoela (400 species): Flatworms in this phylum have a simple nervous system and a saclike gut, and thus were once placed in phylum Platyhelminthes. Recent molecular analyses, however, indicate that Acoela is a separate lineage that diverged before the three main bilaterian clades (see Concept 32.4). Acoela organisms are shown with a scale bar reading 1.5 m m.

Ctenophora (100 species): Ctenophores (comb jellies) are diploblastic and radially symmetrical like cnidarians, suggesting that both phyla diverged early from other animals. Comb jellies make up much of the ocean's plankton. They have eight "combs" of cilia that propel the animals through the water. When a small animal contacts the tentacles of some comb jellies, specialized cells burst open, covering the prey with sticky threads. A photograph of a ctenopbore, or comb jelly is shown.

Lophotrochozoa

- Platyhelminthes (20,000 species): Flatworms in this phylum (including tapeworms, planarians, and flukes) have bilateral symmetry and a central nervous system that processes information from sensory structures. They have no body cavity or specialized organs for circulation (see Concept 33.3). A marine flatworm is shown.
- Syndermata (2,900 species): This recently established phylum includes two groups formerly classified as separate phyla: the rotifers, microscopic animals with complex organ systems, and the acanthocephalans, highly modified parasites of vertebrates (see Concept 33.3). A light microscope image of a rotifer is shown and a scale bar reads 100 micrometers.

- Ectoprocta (4,500 species): Ectoprocts (also known as bryozoans) live as sessile colonies and are covered by a tough exoskeleton (see Concept 33.3). An ectoproct is shown.
- Brachiopoda (335 species): Brachiopods, or lamp shells, may be easily mistaken for clams or other mollusks. However, most brachiopods have a unique stalk that anchors them to their substrate, as well as a crown of cilia called a lophophore (see Concept 33.3). A brachiopod is shown.
- Gastrotricha (800 species): Gastrotrichs are tiny worms whose ventral surface is covered with cilia, leading them to be called hairy bellies. Most species live on the bottoms of lakes or oceans, where they feed on small organ isms and partially decayed organic matter. This individual has consumed algae, visible as the greenish material inside its gut. A differential interference contrast light microscope image of a gastrotrich is shown and a scale bar reads 200 micrometers.
- Cycliophora (1 species): The only known cycliophoran species, Symbion pandora, was discovered in 1995 on the mouthparts of a lobster. This tiny creature has a unique body plan and a bizarre life cycle: Males impregnate females that are still developing in their mothers' bodies. The fertilized females escape, settle elsewhere on the lobster, and release their offspring. A colorized scanning electron microscope image of a cycliophoran is shown and a scale bar reads 100 micrometers.
- Nemertea (900 species): Also called ribbon worms, nemerteans swim through water or burrow in sand, extending a unique proboscis to capture prey. Nemerteans have a reduced coelom, so their bodies are relatively solid. They also have an alimentary canal and a closed circulatory system in which the blood is contained in vessels and hence is distinct from fluid in the body cavity. A photograph of a ribbon worm is shown.
- Annelida (16,500 species): Annelids, or segmented worms, are distinguished by their body segmentation. Earthworms are the most familiar annelids, but most annelids live in marine or freshwater habitats (see Concept 33.3). A photograph of a marine annelid is shown.
- Mollusca (100,000 species): Mollusks (including snails, clams, squids, and octopuses) have a soft body that in many species is protected by a hard shell (see Concept 33.3). A photograph of an octopus is shown.

Ecdysozoa

- Loricifera (10 species): Loriciferans (from the Latin lorica, corset, and ferre, to bear) are tiny animals that inhabit sediments on the seafloor. A loriciferan can telescope its head, neck, and thorax in and out of the lorica, a pocket formed by six plates surrounding the abdomen. Though the natural history of loriciferans is mostly a mystery, at least some species likely eat bacteria. A light microscope image of a loriciferan is shown and a scale bar reads 50 micrometers.
- Priapula (16 species): Priapulans are worms with a large, rounded proboscis at the anterior end. (They are named after Priapos, the Greek god of fertility, who was symbolized by a giant penis.)

Ranging from 0.5 mm to 20 cm in length, most species burrow through seafloor sediments. Fossil evidence suggests that priapulans were among the major predators during the Cambrian period. A photograph of a priapulan is shown.

- Onychophora (110 species): Onychophorans, also called velvet worms, originated during the Cambrian explosion (see Concept 25.3). Originally, they thrived in the ocean, but at some point they succeeded in colonizing land. Today they live only in humid forests. Onychophorans have fleshy antennae and several dozen pairs of saclike legs. A photograph of an onychophoran is shown.
- Tardigrada (800 species): Tardigrades (from the Latin tardus, slow, and gradus, step) are sometimes called water bears for their overall shape and lumbering, bearlike gait. Most tardigrades are less than 0.5 mm in length. Some live in oceans or fresh water, while others live on plants or animals. Harsh conditions may cause tardigrades to enter a state of dormancy; while dormant, they can survive for days at temperatures as low as –200 degrees C! A recent phylogenomic study found that over 15% of tardigrade genes entered their genome by horizontal gene transfer, the largest fraction known for any animal. A scanning electron microscope image of a tardigrade is shown and a scale bar reads 100 micrometers.
- Nematoda (25,000 species): Also called roundworms, nematodes are enormously abundant and diverse in the soil and in aquatic habitats; many species parasitize plants and animals. Their most distinctive feature is a tough cuticle that coats their body (see Concept 33.4). A photograph of a roundworm is shown.
- Arthropoda (1,000,000 species): The vast majority of known animal species, including insects, crustaceans, and arachnids, are arthropods. All arthropods have a segmented exoskeleton and jointed appendages (see Concept 33.4). A photograph of an arachnid, a spider is shown.

Deuterostomia

- Hemichordata (85 species): Like echinoderms and chordates, hemichordates are members of the deuterostome clade (see Figure 32.11). Hemichordates share some traits with chordates, such as gill slits and a dorsal nerve cord. The largest group of hemichordates is the enteropneusts, or acorn worms. Acorn worms are marine and generally live buried in mud or under rocks; they may grow to more than 2 m in length. A photograph of an acorn worm is shown.
- Chordata (60,000 species): More than 90% of all known chordate species have backbones (and thus are vertebrates). However, the phylum Chordata also includes two groups of invertebrates: lancelets and tunicates. See Chapter 34 for a full discussion of this phylum. A photograph of a tunicate is shown.

- Echinodermata (7,000 species): Echinoderms, such as sand dollars, sea stars, and sea urchins, are marine animals in the deuterostome clade that are bilaterally symmetrical as larvae but not as adults. They move and feed by using a network of internal canals to pump water to different parts of their body (see Concept 33.5). A photograph of a sea urchin is shown.

Get Ready for This Chapter

In this chapter, you will learn about the diversity of invertebrates, focusing on the evolutionary history and natural history of these organisms. Before beginning this chapter, you should be able to do the following:

- Describe key characteristics of bilaterian animals (see Concept 32.4).
- Identify a lineage's basal taxon in a phylogenetic tree (see Concept 26.1, Figure 26.5).
- Distinguish among terms related to the body cavity: coelom, hemocoel, acoelomate (see Concept 32.3, Figure 32.9).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements about bilaterian animals is *true*?

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All bilaterians are triploblastic (have three germ layers).
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That's correct.

Correct! If you had trouble with this question, review the following material: **Bilaterian animals** form an enormous clade whose members all have tissues (unlike sponges) as well as a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

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All bilaterians have radial symmetry.
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That's incorrect. Try again.

Incorrect. Review the following material: **Bilaterian animals** form an enormous clade whose members all have a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

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Most bilaterians have tissues, but some do not.
```

That's incorrect. Try again.

Incorrect. Review the following material: **Bilaterian animals** form an enormous clade whose members all have a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

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All bilaterians are invertebrates.
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That's incorrect. Try again.

Incorrect. Review the following material: **Bilaterian animals** form an enormous clade whose members all have a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

Series of nested horizontal forked branches leading to group names; from the top the branch endpoints are Porifera, Ctenophora, Cnidaria, Acoela, Deuterostomia, Lophotrochozoa, and Ecdysozoa.

Diagram showing a series of horizontal forked branches. Reading left to right, the first branch point forks into a horizontal line leading to Porifera and another horizontal line, which branches again. This second branch point forks into a horizontal line leading to Ctenophora and another horizontal line, which branches again. This third branch point forks into a horizontal line leading to Cnidaria and another horizontal line, which branches again. This fourth branch point forks into a horizontal line leading to Acoela and another horizontal line, which branches again. This fifth branch point forks into a horizontal line leading to Deuterostomia and another horizontal line, which branches again into two lines leading to Lophotrochozoa and Ecdysozoa. At the far right of the diagram a vertical bar extending from Ecdysozoa through Acoela is labeled "Bilateria (most animals)." A vertical bar extending from Ecdysozoa through Ctenophora is labeled "Eumetazoa," and another vertical bar extending from Ecdysozoa through Porifera is labeled "Metazoa."

2. Based on the phylogenetic tree shown here, identify the basal taxon of metazoans (animals).

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Ecdysozoa
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That's incorrect. Try again.

Incorrect. Review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that

originates near the common ancestor of the group.

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Eumetazoa
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That's incorrect. Try again.

Incorrect. Review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group.

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Acoela
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That's incorrect. Try again.

Incorrect. Review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group.

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Porifera
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That's correct.

Correct! If you had trouble with this question, review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group. In this phylogenetic tree, Porifera is shown as the first group of metazoans (animals) to have diverged from all other metazoans, and so Porifera is the basal taxon of metazoans.

3. A hemocoel refers to a

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functional body cavity lined by tissues derived from mesoderm and endoderm.
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That's correct.

Correct! If you had trouble with this question, review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from

mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

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        body cavity that does not function properly.
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That's incorrect. Try again.

Incorrect. Review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

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        functional body cavity lined with tissues derived only
from mesoderm.
```

That's incorrect. Try again.

Incorrect. Review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

```
        functional body cavity lined with tissues derived only
from endoderm.
```

That's incorrect. Try again.

Incorrect. Review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 33.1Describe the structure of the sponge body.
- 33.2Characterize cnidarians and their life cycles.
- 33.3Give examples of lophotrochozoans and describe their characteristics.
- 33.4Give examples of ecdysozoans and describe their characteristics.
- 33.5Identify major phylogenetic groups of deuterostomes and describe their characteristics.

Concept 33.1: Sponges are basal animals that lack tissues

An evolutionary tree highlights Porifera.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia.

Animals in the phylum Porifera are known informally as sponges. Among the simplest of animals, sponges are sedentary and were mistaken for plants by the ancient Greeks. Most species are marine, and they range in size from a few millimeters to a few meters. Sponges are **filter feeders**: They filter out food particles suspended in the surrounding water as they draw it through their body, which in some species resembles a sac perforated with pores. Water is drawn through the pores into a central cavity, the **spongocoel**, and then flows out of the sponge through a larger opening called the **osculum(Figure 33.3)**. More complex sponges have folded body walls, and many contain branched water canals and several oscula.

Figure 33.3

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Anatomy of a sponge.
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Image of an azure vase sponge which resembles a vase on the sea floor. Diagram outlines 7 features of basic sponge anatomy.

In the main diagram, portions of the front and back wall are cut away to show the sponge's internal structure.

Seven features of sponge anatomy include:

1.Mesophyll. The wall of this sponge consists of two layers of cells separated by a gelatinous matrix, the mesophyll (middle matter).

2.Epidermis. The outer layer consists of tightly packed epidermal cells.

3.Pores. Water enters the sponge through pores formed by doughnut-shaped cells that span the body wall.

4.Spongocoel. Water passing through pores enters a cavity called the spongocoel.

5.Choanocytes. The spongocoel is lined with flagellated cells called choanocytes. By beating flagella, the choanocytes create a current that draws water in through the pores and out through the osculum.

6.The movement of a choanocyte's flagellum also draws water through its collar of finger-like projections. Food particles are trapped in the mucus that coats the projections, engulfed by phagocytosis, and either digested or transferred to amoebocytes.

7.Amoebocytes. These cells can transport nutrients to other cells of the sponge body, produce materials for skeletal fibers (spicules), or become any type of sponge cell as needed.

Sponges represent a lineage that diverged from other animals early in the history of the group; thus, they are said to be *basal animals*. Unlike nearly all other animals, sponges lack tissues, groups of similar cells that act as a functional unit as in muscle tissue and nervous tissue. However, the sponge body does contain several different cell types. For example, lining the interior of the spongocoel are flagellated **choanocytes**, or collar cells (named for the finger-like projections that form a "collar" around the flagellum). These cells engulf bacteria and other food particles by phagocytosis. The similarity between choanocytes and the cells of choanoflagellates supports molecular evidence indicating that animals and choanoflagellates are sister groups (see Figure 32.3).

The body of a sponge consists of two layers of cells separated by a gelatinous region called the **mesohyl**. Because both cell layers are in contact with water, processes such as gas exchange and waste removal can occur by diffusion across the membranes of these cells. Other tasks are performed by cells called **amoebocytes**, named for their use of pseudopodia. These cells move through the mesohyl and have many functions. For example, they take up food from the surrounding water and from choanocytes, digest it, and carry nutrients to other cells. Amoebocytes also manufacture tough skeletal fibers within the mesohyl. In some sponges, these fibers are sharp spicules made from calcium carbonate or silica. Other sponges produce more flexible fibers composed of a protein called spongin; you may have seen these pliant skeletons being sold as brown bath sponges. Finally, and perhaps most importantly, amoebocytes are *totipotent* (capable of

becoming other types of sponge cells). This gives the sponge body remarkable flexibility, enabling it to adjust its shape in response to changes in its physical environment (such as the direction of water currents).

Most sponges are **hermaphrodites**, meaning that each individual functions as both male and female in sexual reproduction by producing sperm *and* eggs. Almost all sponges exhibit sequential hermaphroditism: They function first as one sex and then as the other. Cross-fertilization can result when sperm released into the water current by an individual functioning as a male is drawn into a neighboring individual that is functioning as a female. The resulting zygotes develop into flagellated, swimming larvae that disperse from the parent sponge. After settling on a suitable substrate, a larva develops into a sessile adult.

Sponges produce a variety of antibiotics and other defensive compounds, which hold promise for fighting human diseases. For example, a compound called cribrostatin isolated from marine sponges can kill both cancer cells and penicillin-resistant strains of the bacterium *Streptococcus*. Other sponge-derived compounds are also being tested as possible anticancer agents.

Concept 33.2: Cnidarians are an ancient phylum of eumetazoans

An evolutionary tree highlights cnidaria, which is the first branch for eumetazoa.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia. Cnidaria, Lophotrochozoa, Ecdysozoa, and Deuterostomia are collectively labeled Eumetazoa.

All animals except sponges and a few other groups are *eumetazoans* (“true animals”), members of a clade of animals with tissues. One of the first lineages to have diverged from others in this clade is the phylum Cnidaria, which originated about 680 million years ago according to DNA analyses. Cnidarians have diversified into a wide range of sessile and motile forms, including hydras, corals, and jellies (commonly called “jellyfish”). Yet most cnidarians still exhibit the relatively simple, diploblastic, radial body plan that existed in early members of the group some 560 million years ago.

The basic body plan of a cnidarian is a sac with a central digestive compartment, the **gastrovascular cavity**. A single opening to this cavity functions as both mouth and anus. There are two variations on this body plan: the largely sessile polyp and the more motile medusa **(Figure 33.4)**. **Polyps** are cylindrical

forms that adhere to the substrate by the aboral end of their body (the end opposite the mouth) and extend their tentacles, waiting for prey. Examples of the polyp form include hydras and sea anemones. Although they are primarily sedentary, many polyps can move slowly across their substrate using muscles at the aboral end of their body. When threatened by a predator, some sea anemones can detach from the substrate and "swim" by bending their body column back and forth, or thrashing their tentacles. A **medusa** (plural, *medusae*) resembles a flattened, mouth-down version of the polyp. It moves freely in the water by a combination of passive drifting and contractions of its bell-shaped body. Medusae include free-swimming jellies. The tentacles of a jelly dangle from the oral surface, which points downward. Some cnidarians exist only as polyps or only as medusae; others have both a polyp stage and a medusa stage in their life cycle.

Figure 33.4

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Polyp and medusa forms of cnidarians.
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Polyp and medusa cross sections.

The body wall of a cnidarian has two layers of cells: an outer layer of epidermis (darker blue; derived from ectoderm) and an inner layer of gastrodermis (yellow; derived from endoderm). Digestion begins in the gastrovascular cavity and is completed inside food vacuoles in the gastrodermal cells. Sandwiched between the epidermis and gastrodermis is a gelatinous layer, the mesoglea.

A polyp has a body stalk that gives it a vase-like shape. A medusa has a jellyfish-like shape with its tentacles below it. Each is mostly surrounded by an epidermis, inside which is the mesoglea. The gastrodermis surrounds the gastrovascular cavity in each. The mouth/anus is at the top of polyp and at the bottom of medusa.

Cnidarians are predators that often use tentacles arranged in a ring around their mouth to capture prey and push the food into their gastrovascular cavity, where digestion begins. Enzymes are secreted into the cavity, thus breaking down the prey into a nutrient-rich broth. Cells lining the cavity then absorb these nutrients and complete the digestive process; any undigested remains are expelled through the cnidarian's mouth/anus. The tentacles are armed with batteries of **cnidocytes**, cells unique to cnidarians that function in defense and prey capture. Cnidocytes contain *cnidae* (from the Greek *cnide*, nettle), capsule-like organelles that are capable of exploding outward and that give phylum Cnidaria its name **(Figure 33.5)**. Specialized cnidae called **nematocysts** contain a stinging thread that can penetrate the body surface of the cnidarian's prey. Other kinds of cnidae have long threads that stick to or entangle small prey that bump into the cnidarian's tentacles.

Figure 33.5

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A cnidocyte of a hydra.
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Diagram zooms in on tip of cnidarian tentacle, lined with cnidocytes containing coiled thread nematocysts with protruding trigger. With a barb-like tip, the thread can puncture body wall of prey.

This type of cnidocyte contains a stinging capsule, the nematocyst, which contains a coiled thread. When a "trigger" is stimulated by touch or by certain chemicals, the thread shoots out, puncturing and injecting poison into prey.

Contractile tissues and nerves occur in their simplest forms in cnidarians. Cells of the epidermis (outer layer) and gastrodermis (inner layer) have bundles of microfilaments arranged into contractile fibers. The gastrovascular cavity acts as a hydrostatic skeleton (see Concept 50.6) against which the contractile cells can work. When a cnidarian closes its mouth, the volume of the cavity is fixed, and contraction of selected cells causes the animal to change shape. Cnidarians have no brain. Instead, movements are coordinated by a noncentralized *nerve net* that is associated with sensory structures distributed around the body. Thus, even though it lacks a brain, the animal can detect and respond to stimuli from all directions.

Fossil and molecular evidence suggests that early in its evolutionary history, the phylum Cnidaria diverged into two major clades, Medusozoa and Anthozoa **(Figure 33.6)**.

Figure 33.6

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Cnidarians.
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Cnidarians are described.

Part a. Medusozoans shown include jellies and sea wasps. Jellies: Many jellies are bioluminescent. Food captured by nematocyst-bearing tentacles is transferred to specialized oral arms (that lack nematocysts) for transport to the mouth. Jelly pictured is translucent with dome-like top and long tentacles. Jellies here glow against a dark background. Sea wasps: This sea wasp produces a poison that can subdue fish, crustaceans, and other large prey. The poison is more potent than cobra venom. The sea wasp, or box jelly pictured, attacks prey with a tentacle.

Part b. Anthozoans shown include sea anemones and star corals. Sea anemones: Sea anemones and other anthozoans exist only as polyps. Many anthozoans form symbiotic relationships with photosynthetic algae. Star corals: These star corals live as colonies of polyps. Their soft bodies are enclosed at the base by a hard exoskeleton. Photograph of a sea anemone with short, stalk-like body topped with tentacles. Star corals appear as a cluster of smaller polyps that resemble anemones.

Video

Jelly Swimming

image

image

Video

Hydra Eating *Daphnia*

image

image

Medusozoans

All cnidarians that produce a medusa are members of clade Medusozoa, a group that includes the *scyphozoans* (jellies) and *cubozoans* (box jellies) shown in Figure 33.6a, along with the *hydrozoans*. Most hydrozoans alternate between the polyp and medusa forms, as seen in the life cycle of *Obelia***(Figure 33.7)**. The polyp stage, a colony of interconnected polyps in the case of *Obelia*, is more conspicuous than the medusa. Hydras, among the few cnidarians found in fresh water, are also unusual hydrozoans in that they exist only in polyp form.

Figure 33.7

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The life cycle of the hydrozoan *Obelia*.
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Life cycle diagram with inset photograph highlights the processes of meiosis and fertilization, as well as haploid and diploid stages.

The polyp is asexual, and the medusa is sexual, releasing eggs and sperm. These two stages alternate, one producing the other.

The cycle is as follows.

1.A colony of interconnected polyps (diploid) results from asexual reproduction by budding. Colony resembles a long stalk with bud-like structures branching from it, some with fringed ends and some with smooth ends. Scale for colony image = 1 millimeter.

2.Some of the colony's (diploid) polyps, equipped with tentacles (the fringed ends) are specialized for feeding.

3.Other (diploid) polyps, specialized for reproduction, lack tentacles (have smooth ends) and produce tiny medusae (diploid) by asexual budding.

4.Diploid medusae, which are dome-shaped with tentacle fringes and gonads under the domes, swim off, grow, and reproduce sexually. Following meiosis, some medusae produce eggs (haploid) from their gonads and some produce sperm (haploid). Haploid sperm fertilize haploid eggs, forming diploid zygotes, which begin to divide.

5.The diploid zygotes develop into a solid ciliated larva called a planula (diploid). Diagram shows a rod-shaped organism surrounded by a fringe of cilia.

6.The planula eventually settles and develops a new (diploid) polyp. The polyp matures and the cycle can begin again at step 1.

MAKE CONNECTIONS *Compare and contrast the* Obelia *life cycle to the life cycles in*Figure 13.6*.* Which life cycle in that figure is most similar to that of* Obelia? *Explain. (See also*Figure 29.5*.)**

*For suggested answer, see*Appendix A*.**

Unlike hydrozoans, most scyphozoans and cubozoans spend the majority of their life cycles in the medusa stage. Coastal scyphozoans, for example, often have a brief polyp stage during their life cycle, whereas those that live in the open ocean generally lack the polyp stage altogether. As their name (which means "cube animals") suggests, cubozoans have a box-shaped medusa stage. Most cubozoans live in tropical oceans and are equipped with highly toxic cnidocytes. For example, the sea wasp (*Chironex fleckeri*), a cubozoan that lives off the coast of northern Australia, is one of the deadliest organisms known: Its sting causes intense pain and can lead to respiratory failure, cardiac arrest, and death within minutes.

Anthozoans

Sea anemones and corals belong to the clade Anthozoa (see Figure 33.6b). These cnidarians occur only as polyps. Corals live as solitary or colonial forms, often forming symbioses with algae. Many species secrete a hard **exoskeleton** (external skeleton) of calcium carbonate. Each polyp generation builds on the skeletal remains of earlier generations, constructing rocklike reefs with shapes characteristic of their species. These skeletons are what we usually think of as coral.

Coral reefs are to tropical seas what rain forests are to tropical land areas: They provide habitat for many other species. Unfortunately, these reefs are being destroyed at an alarming rate. Pollution, overharvesting,

and ocean acidification (see Figure 3.12) are major threats. Climate change is also contributing to their demise by raising seawater temperatures above the range in which corals thrive (see Concept 56.4).

HHMI Biointeractive Video

Coral Bleaching

image

image

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Concept 33.3: Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms

An evolutionary tree highlights lophotrochozoa, which is the first branch for bilateria.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia. Lophotrochozoa, Ecdysozoa, and Deuterostomia are collectively labeled Bilateria.

The vast majority of animal species belong to the clade Bilateria, whose members exhibit bilateral symmetry and are triploblastic (see Concept 32.3). Most bilaterians also have a digestive tract with two openings (a mouth and an anus) and a body cavity (a coelom or hemocoel; see Figure 32.9). Recent DNA analyses suggest that the common ancestor of living bilaterians lived about 670millionyears ago. To date, however, the oldest fossil that is widely accepted as a bilaterian is of *Kimberella*, a mollusc (or close relative) that lived 560millionyears ago (see Figure 32.5b). Many other bilaterian groups first appeared in the fossil record during the Cambrian explosion (535 to 525 million years ago).

Molecular evidence suggests that today there are three major clades of bilaterally symmetrical animals: Lophotrochozoa, Ecdysozoa, and Deuterostomia. This section will focus on the first of these clades, the lophotrochozoans. Concepts 33.4 and 33.5 will explore the other two clades.

Although the clade Lophotrochozoa was identified by molecular data, its name comes from features found in some of its members. Some lophotrochozoans develop a structure called a *lophophore*, a crown of ciliated tentacles that functions in feeding, while others go through a distinctive stage called the *trochophore larva* (see Figure 32.12). Other members of the group have neither of these features. Few other unique morphological features are widely shared within the group—in fact, the lophotrochozoans are the most diverse bilaterian clade in terms of body plan. This diversity in form is reflected in the number of phyla classified in the group: Lophotrochozoa includes 18 phyla, more than twice the number in any other clade of bilaterians.

We'll now introduce six of the diverse lophotrochozoan phyla: the flatworms, rotifers and acanthocephalans, ectoprocts, brachiopods, molluscs, and annelids.

Flatworms

Flatworms (phylum Platyhelminthes) live in marine, freshwater, and damp terrestrial habitats. In addition to free-living species, flatworms include many parasitic species, such as flukes and tapeworms. Flatworms are so named because they have thin bodies that are flattened dorsoventrally (between the dorsal and ventral surfaces); the word *platyhelminth* means "flat worm." (Note that *worm* is not a formal scientific name but rather refers to animals with long, thin bodies.) The smallest flatworms are nearly microscopic free-living species, while some tapeworms are more than 20 m long.

Although flatworms are triploblastic, they lack a body cavity. Their flat shape increases their surface area, placing all their cells close to water in the surrounding environment or in their gut. Because of this proximity to water, gas exchange and the elimination of nitrogenous waste (ammonia) can occur by diffusion across the body surface. As shown in **Figure 33.8**, a flat shape is one of several structural features that maximize surface area and have arisen (by convergent evolution) in different groups of animals and other organisms.

Figure 33.8

AP Make Connections: Maximizing Surface Area

Make connections: Maximizing surface area.

The diagram reads as follows.

• Maximizing surface area: In general, the amount of metabolic or chemical activity an organism can carry out is proportional to its mass or volume. Maximizing metabolic rate, however, requires the efficient uptake of energy and raw materials, such as nutrients and oxygen, as well as the effective disposal of waste products. For large cells, plants, and animals, these exchange processes have the potential to be limiting due to simple

geometry. When a cell or organism grows without changing shape, its volume increases more rapidly than its surface area (see figure 6.7). As a result, there is proportionately less surface area over which exchange processes can occur. The challenge posed by the relationship of surface area and volume occurs in diverse contexts and organisms, but the evolutionary adaptations that meet this challenge are similar. Structures that maximize surface area through flattening, folding, branching, and projections have an essential role in biological systems.

• Flattening: By having a body that is only a few cells thick, an organism such as a flatworm can use its entire body surface for exchange. (See figure 40.3).

• Folding: A TEM shows portions of two chloroplasts in a plant leaf. Photosynthesis occurs in chloroplasts, which have a flattened and interconnected set of internal membranes called thylakoid membranes. The foldings of the thylakoid membranes increase their surface area, enhancing the exposure to light and thus increasing the rate of photosynthesis. (See figure 10.4).

• Branching: Water uptake relies on passive diffusion. The highly branched filaments of a fungal mycelium increase the surface area across which water and minerals can be absorbed from the environment. (See figure 31.2).

• Projections: In vertebrates, the small intestine is lined with finger-like projections called villi that absorb nutrients released by the digestion of food. Each of the villi shown here is covered with large numbers of microscopic projections called microvilli, resulting in a total surface area of about 300 m^2 in humans, as large as a tennis court. (See figure 41.12).

Diagram compares surface area to volume for two different shapes with the same volume. Both shapes are subdivided into smaller squares of equal size for reference. The first shape is a cube that is 3 one centimeter squares high and 3 across (for a total of nine centimeter squares cubes making up each side of the cube).

Calculation for surface area of the cube: 6 times (3 centimeters times 3 centimeters) = 54 centimeters squared.

Volume of the cube: 3 centimeters times 3 centimeters times 3 centimeters = 27 centimeters cubed.

The second shape is a flattened rectangle that is 3-one centimeter squares wide, one centimeter square in height and 9 centimeter squares long.

Calculation of the surface area in the rectangle: 2 times (3 centimeters times 1 centimeter) + 2 times (9 centimeters times 1 centimeter) + 2 times (3 centimeters times 9 centimeters) = 78 centimeters squared.

Volume: 1 centimeter times 3 centimeters times 9 centimeters = 27 centimeters cubed.

Thus, the cube has a surface area to volume ratio of 54 to 27, while the flattened rectangle has a surface area to volume ratio of 78 to 27. This demonstrates that the flattened shape has a much higher surface area relative to its volume than the fatter cube shape.

MAKE CONNECTIONS *Find other examples of flattening, folding, branching, and projections (see*Chapters 6*,***9**,**35**, and**42**). How is maximizing surface area important to the structure's function in each example?*

*For suggested answer, see*Appendix A.*

As you might expect since all their cells are close to water, flatworms have no organs specialized for gas exchange, and their relatively simple excretory apparatus functions mainly to maintain osmotic balance with their surroundings. This apparatus consists of **protonephridia**, networks of tubules with ciliated structures called *flame bulbs* that pull fluid through branched ducts opening to the outside (see Figure 44.8). Most flatworms have a gastrovascular cavity with only one opening. Though flatworms lack a circulatory system, the fine branches of the gastrovascular cavity distribute food directly to the animal's cells.

Early in their evolutionary history, flatworms separated into two lineages, Catenulida and Rhabditophora. Catenulida is a small clade of about 100 flatworm species, most of which live in freshwater habitats. Catenulids typically reproduce asexually by budding at their posterior end. The offspring often produce their own buds before detaching from the parent, thereby forming a chain of two to four genetically identical individuals—hence their informal name, "chain worms."

The other ancient flatworm lineage, Rhabditophora, is a diverse clade of about 20,000 freshwater and marine species, one example of which is shown in Figure 33.8. We'll explore the rhabditophorans in more detail, focusing on free-living and parasitic members of this clade.

Free-Living Species

Free-living rhabditophorans are important as predators and scavengers in a wide range of freshwater and marine habitats. The best-known members of this group are freshwater species in the genus *Dugesia*, commonly called **planarians**. Abundant in unpolluted ponds and streams, planarians prey on smaller animals or feed on dead animals. They move by using cilia on their ventral surface, gliding along a film of mucus they secrete. Some other rhabditophorans also use their muscles to swim through water with an undulating motion.

A planarian's head features a pair of light-sensitive eyespots as well as lateral flaps that function mainly to detect specific chemicals. The planarian nervous system is more complex and centralized than the nerve nets of cnidarians **(Figure 33.9)**. Experiments have shown that planarians can learn to modify their responses to stimuli.

Figure 33.9

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Anatomy of a planarian.
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Planarian anatomy detailed, with the functions of pharynx, ganglia, ventral nerve cords and gastrovascular cavity. The planarian has a flat body with a roughly triangular anterior end containing two eyespots.

Digestion is completed within the cells lining the gastrovascular cavity, which has many fine subbranches that provide an extensive surface area. Undigested wastes are egested through an opening at the tip of the pharynx.

Ventral nerve cords. From the ganglia, a pair of ventral nerve cords runs the length of the body.

Pharynx. A muscular pharynx can be extended through the mouth. Digestive juices are spilled onto prey, and the pharynx sucks small pieces of food into the gastrovascular cavity, where digestion continues.

Ganglia. At the anterior end of the worm, near the main sources of sensory input, is a pair of ganglia, dense clusters of nerve cells. Body shape is flat. The anterior end is roughly triangular and contains two eyespots on the dorsal side as well as two nearby ganglia (one associated with each eyespot). The gastrovascular cavity lines the middle of the interior of the organism. Near the middle of the body, on the ventral side, is the mouth opening, through which the pharynx can extend (resembles a tube).

Some planarians can reproduce asexually through fission. The parent constricts roughly in the middle of its body, separating into a head end and a tail end; each end then regenerates the missing parts. Sexual reproduction also occurs. Planarians are hermaphrodites, and copulating mates typically cross-fertilize each other.

Parasitic Species

More than half of the known species of rhabditophorans live as parasites in or on other animals. Many have suckers that attach to the internal organs or outer surfaces of the host animal. In most species, a tough covering helps protect the parasites within their hosts. We'll discuss two ecologically and economically important subgroups of parasitic rhabditophorans, the trematodes and the tapeworms.

Trematodes

As a group, trematodes parasitize a wide range of hosts, and most species have complex life cycles with alternating sexual and asexual stages. Many trematodes require an intermediate host in which larvae develop before infecting the final host (usually a vertebrate), where the adult worms live. For example,

various trematodes that parasitize humans spend part of their lives in snail hosts **(Figure 33.10)**. Around the world, about 200million people are infected with trematodes called blood flukes (*Schistosoma*) and suffer from schistosomiasis, a disease whose symptoms include pain, anemia, and diarrhea.

Figure 33.10

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    The life cycle of a blood fluke (*Schistosoma mansoni*), a
trematode.
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Lifecycle diagram depicts various stages and different hosts (human, snail) for a trematode.

The cycles is as follows.

1.Mature flukes live in the blood vessels of the human intestine (person drawn here pulling plants from shallow water). A female fluke fits into a groove running the length of the larger male's body (male appears wormlike in LM, scale 1 millimeter); female is a darker and smaller worm-like organism within the male.

2.Blood flukes reproduce sexually in the human host. The fertilized eggs exit the host in the feces.

3.If the human feces reach a pond or other source of water (as they do in this diagram) then the eggs develop into ciliated larvae (organism with cilia that appear as small hairs on the outer surface). These larvae infect snails, the intermediate host.

4.Asexual reproduction within a snail results in another type of motile larva, which escapes from the snail host. This larva is drawn as a longer than the larva in step 3, also with cilia and a branched tail.

5.These larvae penetrate the skin and blood vessels of humans working in fields irrigated with water contaminated with fluke larvae (and cycle can begin again at step 1).

WHAT IF? *Snails eat algae, whose growth is stimulated by nutrients found in fertilizer. How would the contamination of irrigation water with fertilizer likely affect the occurrence of schistosomiasis? Explain.*

*For suggested answer, see*Appendix A.**

Living within more than one kind of host puts demands on trematodes that free-living animals don't face. A blood fluke, for instance, must evade the immune systems of both snails and humans. By mimicking the surface proteins of its hosts, the blood fluke creates a partial immunological camouflage for itself. It also releases molecules that manipulate the hosts' immune systems into tolerating the parasite's existence. These defenses are so effective that individual blood flukes can survive in humans for more than 40 years.

Tapeworms

The tapeworms are a second large and diverse group of parasitic rhabditophorans **(Figure 33.11)**. The adults live mostly inside vertebrates, including humans. In many tapeworms, the anterior end, or *scolex*, is armed with suckers and often hooks that the worm uses to attach itself to the intestinal lining of its host. Tapeworms lack a mouth and gastrovascular cavity; they simply absorb nutrients released by digestion in the host's intestine. Absorption occurs across the tapeworm's body surface.

Figure 33.11

Anatomy of a tapeworm.

Tapeworm body is long, ribbon-like and lined with proglottids with reproductive structures. The scolex, at one end, is seen in a close-up image; contains hooks and suckers. Scale: 100 micrometers.

The inset shows a close-up of the scolex (colorized SEM).

Posterior to the scolex is a long ribbon of units called *proglottids*, which are little more than sacs of sex organs. After sexual reproduction, proglottids loaded with thousands of fertilized eggs are released from the posterior end of a tapeworm and leave the host's body in feces. In one type of life cycle, feces carrying the eggs contaminate the food or water of intermediate hosts, such as pigs or cattle, and the tapeworm eggs develop into larvae that encyst in muscles of these animals. A human acquires the larvae by eating undercooked meat containing the cysts, and the worms develop into mature adults within the human. Large tapeworms can block the intestines and rob enough nutrients from the human host to cause nutritional deficiencies. Several different oral medications can kill the adult worms.

Rotifers and Acanthocephalans

Recent phylogenetic analyses have shown that two traditional animal phyla, the rotifers (former phylum Rotifera) and the acanthocephalans (former phylum Acanthocephala), should be combined into a single phylum, Syndermata. Each of the two groups has distinctive characteristics.

Rotifers

There are roughly 1,800 species of rotifers, tiny animals that inhabit freshwater, marine, and damp soil habitats. Ranging in size from about 50 μm to 2 mm, rotifers are smaller than many protists but nevertheless are multicellular and have specialized organ systems **(Figure 33.12)**. In contrast to cnidarians and flatworms, which have a gastrovascular cavity, rotifers have an **alimentary canal**, a digestive tube with two openings, a mouth and an anus. Internal organs lie within the *hemocoel* (see Figure 32.9b). Fluid in the hemocoel serves as a hydrostatic skeleton. Movement of a rotifer's body distributes the fluid throughout the body, circulating nutrients.

Figure 33.12

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A rotifer.
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Rotifer: Wine glass-shaped organism with crown of cilia around mouth with jaws behind, stomach in the center and anus at the base near what would be the wine glass stem. Scale: 0.1 millimeters.

Although smaller than many unicellular protists, rotifers are multicellular and have specialized organ systems (LM).

The word *rotifer* is derived from the Latin meaning “wheel-bearer,” a reference to the crown of cilia that draws a vortex of water into the mouth. Posterior to the mouth, rotifers have jaws called trophi that grind up food, mostly microorganisms suspended in the water. Digestion is then completed farther along the alimentary canal. Most other bilaterians also have an alimentary canal, which enables the stepwise digestion of a wide range of food particles.

Rotifers exhibit some unusual forms of reproduction. Some species consist only of females that produce more females from unfertilized eggs, a type of asexual reproduction called **parthenogenesis**. Some other invertebrates (for example, aphids and some bees) and even some vertebrates (for example, some lizards and some fishes) can also reproduce in this way. In addition to being able to produce females by parthenogenesis, some rotifers can also reproduce sexually under certain conditions, such as high levels of crowding. The resulting embryos can remain dormant for years. Once they break dormancy, the embryos develop into another generation of females that reproduce asexually.

It is puzzling that many rotifer species persist without males. The vast majority of animals and plants reproduce sexually at least some of the time, and sexual reproduction has certain advantages over asexual reproduction (see Concept 46.1). For example, species that reproduce asexually tend to accumulate harmful mutations in their genomes faster than sexually reproducing species. As a result, asexual species should experience higher rates of extinction.

Seeking to understand how they persist without males, researchers have been studying a clade of asexual rotifers named Bdelloidea. Some 360 species of bdelloid rotifers are known, and all of them reproduce by parthenogenesis without any males. Paleontologists have discovered bdelloid rotifers preserved in 35-million-year-old amber, and the morphology of these fossils resembles only the female form, with no evidence of males. Molecular clock analyses indicate that bdelloids have been asexual for over 50million years. While it appears that they do not reproduce sexually, bdelloid rotifers may be able to generate genetic diversity in other ways. For example, bdelloids can tolerate very high levels of desiccation. When conditions improve and their cells rehydrate, DNA from other species enters their cells through cracks in the plasma

membrane. Recent evidence suggests that this foreign DNA can be incorporated into the bdelloids' genome, thereby leading to increased genetic diversity.

Acanthocephalans

Acanthocephalans (1,100 species) are sexually reproducing parasites of vertebrates that lack a complete digestive tract and usually are less than 20cm long. They are commonly called spiny-headed worms because of the curved hooks on the proboscis at the anterior end of their body **(Figure 33.13)**. Acanthocephalans are triploblastic and, like rotifers, they have a hemocoel. Recent studies have shown that acanthocephalans originated from within the group traditionally known as Rotifera. In particular, rotifers in the genus *Seison* share a more recent common ancestor with acanthocephalans than they do with other rotifers, making the acanthocephalans a group of highly modified "rotifers."

Figure 33.13

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*Paratenuisentis ambiguus,* an acanthocephalan.
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Photo of Paratenuisentis ambiguus, shaped like a thin crescent, with close-up of the proboscis, which looks like a cactus with curved spines.

The inset photograph shows the curved hooks that give the spiny-headed worms their name.

All acanthocephalans are parasites that have complex life cycles with two or more hosts. Some species manipulate the behavior of their intermediate hosts (generally arthropods) in ways that increase their chances of reaching their final hosts (generally vertebrates). For example, acanthocephalans that infect New Zealand mud crabs cause their hosts to move to more visible areas on the beach, where the crabs are more likely to be eaten by birds, the worms' final hosts.

Ectoprocts and Brachiopods

Bilaterians in the phyla Ectoprocta and Brachiopoda have a *lophophore*, a crown of ciliated tentacles around their mouth (see Figure 32.12a). As the cilia draw water toward the mouth, the tentacles trap suspended food particles. Other similarities, such as a U-shaped alimentary canal and the absence of a distinct head, reflect these organisms' sessile existence. In contrast to flatworms, which lack a body cavity, and rotifers, which have a hemocoel, ectoprocts and brachiopods have a coelom (see Figure 32.9a).

Ectoprocts (from the Greek *ecto*, outside, and *procta*, anus) are colonial animals that superficially resemble clumps of moss. (In fact, their common name, bryozoans, means "moss animals.") In most species,

the colony is encased in a hard exoskeleton studded with pores through which the lophophores extend **(Figure 33.14a)**. Most ectoproct species live in the sea, where they are among the most widespread and numerous sessile animals. Several species are important reef builders. Ectoprocts also live in lakes and rivers. Colonies of the freshwater ectoproct *Pectinatella magnifica* grow on submerged sticks or rocks and can grow into a gelatinous, ball-shaped mass more than 10cm across.

Figure 33.14

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Ectoprocts and brachiopods.
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Two lophophorates.

Part a. Image of Ectoproct with stalk-like structure through which several fan-like lipophores extend.

Part b. Image of brachiopod with hinged shell through which fan-like lophophore extends. The two parts of the shell are dorsal and ventral.

Brachiopods, or lamp shells, superficially resemble clams and other hinge-shelled molluscs, but the two halves of the brachiopod shell are dorsal and ventral rather than lateral, as in clams **(Figure 33.14b)**. All brachiopods are marine. Most live attached to the seafloor by a stalk, opening their shell slightly to allow water to flow through the lophophore. The living brachiopods are remnants of a much richer past that included 30,000 species in the Paleozoic and Mesozoic eras. Some living brachiopods, such as those in the genus *Lingula*, appear nearly identical to fossils of species that lived 400millionyears ago.

Molluscs

Snails and slugs, oysters and clams, and octopuses and squids are all molluscs (phylum Mollusca). There are over 100,000 known species, making them the second most diverse phylum of animals (after the arthropods, discussed later). Although the majority of molluscs are marine, roughly 8,000 species inhabit fresh water, and 28,000 species of snails and slugs live on land. All molluscs are soft-bodied, and most secrete a hard protective shell made of calcium carbonate. Slugs, squids, and octopuses have a reduced internal shell or have lost their shell completely during their evolution.

Despite their apparent differences, all molluscs have a similar body plan **(Figure 33.15)**. The primary body cavity is a hemocoel, but they also have a reduced coelom. The body of a mollusc has three main parts: a muscular **foot**, usually used for movement; a **visceral mass** containing most of the internal organs; and a **mantle**, a fold of tissue that drapes over the visceral mass and secretes a shell (if one is present). In many molluscs, the mantle extends beyond the visceral mass, producing a water-filled chamber, the **mantle**

cavity, which houses the gills, anus, and excretory pores. Many molluscs feed by using a straplike organ called a **radula** to scrape up food.

Figure 33.15

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The basic body plan of a mollusc.
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Diagram of a mollusc body plan shows a domed-shape with shell and two openings over the foot and many different organs, with some functions described.

Metanephridium. Excretory organs called metanephridia remove metabolic wastes from the hemolymph. Heart. Most mollusks have an open circulatory system. The dorsally located heart pumps circulatory fluid called hemolymph through arteries into sinuses (body spaces). The organs of the mollusc are thus continually bathed in hemolymph. The long digestive tract is coiled in the visceral mass.

Radula. The mouth region in many mollusc species contains a rasp-like feeding organ called a radula. This belt of backward-curved teeth repeatedly thrusts outward and then retracts into the mouth, scraping and scooping like a backhoe. The nervous system consists of a nerve ring around the esophagus, from which nerve cords extend. Basic body plan shown here is a dome-shaped shell, beneath which is the mantle and mantle cavity, followed by the visceral mass containing most of the organs. The foot is below (ventral to) the visceral mass and has branched nerve cords running through it, with one end of the nerve cord forming a ring around the esophagus. At the anterior end is the mouth region containing the radula, with a blade of backwards-curving teeth. At the posterior end is the anus. Other organs detailed in the visceral mass, from roughly anterior to posterior, are the esophagus (just posterior to the radula), and the stomach and intestines, which connect to the anus. Above (dorsal to) the stomach and intestines are the gonads followed by the coelom and the metanephridium and the heart, which is connected to the feather-like gill at the posterior of the body.

Most molluscs have separate sexes, and their gonads (ovaries or testes) are located in the visceral mass. Many snails, however, are hermaphrodites. The life cycle of many marine molluscs includes a ciliated larval stage, the trochophore (see **Figure 32.12b**), which is also characteristic of marine annelids (segmented worms) and some other lophotrochozoans.

The basic body plan of molluscs has evolved in various ways in the phylum's eight major clades. We'll examine four of those clades here: Polyplacophora (chitons), Gastropoda (snails and slugs), Bivalvia (clams, oysters, and other bivalves), and Cephalopoda (squids, octopuses, cuttlefishes, and chambered nautiluses). We will then focus on threats facing some groups of molluscs.

Chitons

Chitons have an oval-shaped body and a shell composed of eight dorsal plates **(Figure 33.16)**. The chiton's body itself, however, is unsegmented. You can find these marine animals clinging to rocks along the shore during low tide. If you try to dislodge a chiton by hand, you will be surprised at how well its foot, acting as a suction cup, grips the rock. A chiton can also use its foot to creep slowly over the rock surface. Chitons use their radula to scrape algae off the rock surface.

Figure 33.16

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A chiton.
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Image of chiton shows oval-shaped shell divided into 8 plate segments on a rock under water.

Note the eight-plate shell characteristic of molluscs in the clade Polyplacophora.

Gastropods

About three-quarters of all living species of molluscs are gastropods **(Figure 33.17)**. Most gastropods are marine, but there are also freshwater species. Still other gastropods have adapted to life on land, where snails and slugs thrive in habitats ranging from deserts to rain forests.

Figure 33.17

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Gastropods.
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Part a. Land snail has a cylindrical body with antenna on head and spiraling shell. Part b. Sea slug has a cylindrical body and bumpy-looking covering on dorsal side in place of shell.

The many species of gastropods have colonized terrestrial as well as aquatic environments.

Gastropods move literally at a snail's pace by a rippling motion of their foot or by means of cilia—a slow process that can leave them vulnerable to attack. Most gastropods have a single, spiraled shell into which the animal can retreat when threatened. The shell, which is secreted by glands at the edge of the mantle, has several functions, including protecting the animal's soft body from injury and dehydration. One of its most important roles is as a defense against predators, as is demonstrated by comparing populations with different histories of predation (see the **Scientific Skills Exercise**). As they move slowly about, most gastropods use their radula to graze on algae or plants. Several groups, however, are predators, and their

radula has become modified for boring holes in the shells of other molluscs or for tearing apart prey. In the cone snails, the teeth of the radula act as poison darts that are used to subdue prey.

Scientific Skills Exercise

Understanding Experimental Design and Interpreting Data

AP Science Practices 3, 4, 6

Is There Evidence of Selection for Defensive Adaptations in Mollusc Populations Exposed to Predators? The fossil record shows that historically, increased risk to prey species from predators is often accompanied by increased incidence and expression of prey defenses. Researchers tested whether populations of the predatory European green crab (*Carcinus maenas*) have exerted similar selective pressures on its gastropod prey, the flat periwinkle (*Littorina obtusata*). Periwinkles from southern sites in the Gulf of Maine have experienced predation by European green crabs for over 100 generations, at about one generation per year. Periwinkles from northern sites in the Gulf have been interacting with the invasive green crabs for relatively few generations, as the invasive crabs spread to the northern Gulf comparatively recently.

A periwinkle is a small domed shell with a soft-bodied organism beneath.

A periwinkle

Previous research shows that (1) periwinkle shells recently collected from the Gulf are thicker than those collected in the late 1800s, and (2) periwinkle populations from southern sites have thicker shells than periwinkle populations from northern sites. In this exercise, you'll interpret the design and results of the researchers' experiment studying the rates of predation by crabs on periwinkles from northern and southern populations.

How the Experiment Was Done The researchers collected periwinkles and crabs from sites in the northern and southern Gulf of Maine, separated by 450 km of coastline. A single crab was placed in a cage with eight periwinkles of different sizes. After three days, researchers assessed the fate of the eight periwinkles. Four different treatments were set up, with crabs from northern or southern populations offered periwinkles from northern and southern populations. All crabs were of similar size and included equal numbers of males and females. Each experimental treatment was tested 12 to 14 times.

In a second part of the experiment, the bodies of periwinkles from northern and southern populations were removed from their shells and presented to crabs from northern and southern populations.

Data from the Experiment

Bar graph shows average numbers of northern and southern periwinkles killed by crabs. At least 5 times as many northern as southern periwinkles were killed regardless of crab source.

Data from R. Rochette et al., Interaction between an invasive decapod and a native gastropod: Predator foraging tactics and prey architectural defenses, *Marine Ecology Progress Series* 330:179–188 (2007).

One bar represents northern periwinkles and one represents southern periwinkles. For crabs from southern population sources, an average of a little more than 1 southern periwinkle was killed; for the same crab source population, a little more than 5 northern periwinkles were killed. For crabs from northern source populations, average southern periwinkles killed were just above zero; for the same crab source population, an average of over 4 northern periwinkles were killed.

When the researchers presented the crabs with unshelled periwinkles, all the unshelled periwinkles were consumed in less than an hour.

INTERPRET THE DATA

- What hypotheses were the researchers testing in this study? What are the independent variables? What are the dependent variables?
- Why did the research team set up four different treatments?
- Why did researchers present unshelled periwinkles to the crabs? What do the results of this part of the experiment indicate?
- Summarize the results of the experiment. Do these results support the hypothesis you identified in question 1? Explain.
- Suggest how natural selection may have affected populations of flat periwinkles in the southern Gulf of Maine over the last 100 years.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Many gastropods have a head with eyes at the tips of tentacles. Terrestrial snails lack the gills typical of most aquatic gastropods. Instead, the lining of their mantle cavity functions as a lung, exchanging respiratory gases with the air.

Bivalves

The molluscs of the clade Bivalvia are all aquatic and include many species of clams, oysters, mussels, and scallops. Bivalves have a shell divided into two halves **(Figure 33.18)**. The halves are hinged, and powerful adductor muscles draw them tightly together to protect the animal's soft body. Bivalves have no distinct head, and the radula has been lost. Some bivalves have eyes and sensory tentacles along the outer edge of their mantle.

Figure 33.18

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A bivalve.
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A bivalve with two scallop-edged shells that hinge together, seen on a sand bed.

This scallop has many eyes (dark blue spots) lining each half of its hinged shell.

The gills of bivalves are used for feeding as well as gas exchange in most species **(Figure 33.19)**. Most bivalves are suspension feeders. They trap small food particles in mucus that coats their gills, and cilia then convey those particles to the mouth. Water enters the mantle cavity through an incurrent siphon, passes over the gills, and then exits the mantle cavity through an excurrent siphon.

Figure 33.19

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Anatomy of a clam.
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Organs and structures in a diagram of a clam are labeled.

Food particles suspended in water that enters through the incurrent siphon are collected by the gills and passed via cilia and the palps to the mouth.

Shell is roughly heart-shaped with a hinge at one end. Just inside the shell are the mantle and mantle cavity. Opposite the hinge area is the opening that is the incurrent siphon (water flows in); adjacent is the excurrent siphon (water flows out after passing over the gills). Near the excurrent siphon is the anus; adjacent are one or two adductor muscles. The anus is connected to the heart, gut, palp, digestive gland and mouth (opposite the anus). The foot is a large organ within the mantle cavity; also present are the gonad and coelom.

Most bivalves lead sedentary lives, a characteristic suited to suspension feeding. Mussels secrete strong threads that tether them to rocks, docks, boats, and the shells of other animals. However, clams can pull themselves into the sand or mud, using their muscular foot for an anchor, and scallops can skitter along the seafloor by flapping their shells, rather like the mechanical false teeth sold in novelty shops.

Cephalopods

Cephalopods are active marine predators **(Figure 33.20)**. They use their tentacles to grasp prey, which they then bite with beak-like jaws and immobilize with a poison present in their saliva. The foot of a cephalopod has become modified into a muscular excurrent siphon and part of the tentacles. Squids dart about by drawing water into their mantle cavity and then firing a jet of water through the excurrent siphon; they steer by pointing the siphon in different directions. Octopuses use a similar mechanism to escape predators.

Figure 33.20

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Cephalopods.
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Cephalopods. Squids are speedy carnivores with beak-like jaws and well-developed eyes. Octopuses are considered among the most intelligent invertebrates. Chambered nautiluses are the only living cephalopods with an external shell.

The mantle covers the visceral mass of cephalopods, but the shell is generally reduced and internal (in most species) or missing altogether (in some cuttlefishes and some octopuses). One small group of cephalopods with external shells, the chambered nautiluses, survives today.

Cephalopods are the only molluscs with a *closed circulatory system*, in which the blood remains separate from fluid in the body cavity. They also have well-developed sense organs and a complex brain. The ability to learn and behave in a complex manner is probably more critical to fast-moving predators than to sedentary animals such as clams.

The ancestors of octopuses and squids were probably shelled molluscs that took up a predatory lifestyle; the shell was lost in later evolution. Shelled cephalopods called **ammonites**, some of them as large as truck tires, were the dominant invertebrate predators of the seas for hundreds of millions of years until their disappearance during the mass extinction at the end of the Cretaceous period, 66 million years ago.

Most species of squid are less than 75 cm long, but some are much larger. The giant squid (*Architeuthis dux*), for example, has an estimated maximum length of 13 m for females and 10 m for males. The colossal squid (*Mesonychoteuthis hamiltoni*), is even larger, with an estimated maximum length of 14 m. Unlike *A. dux*, which has large suckers and small teeth on its tentacles, *M. hamiltoni* has two rows of sharp hooks at the ends of its tentacles that can inflict deadly lacerations.

It is likely that *A. dux* and *M. hamiltoni* spend most of their time in the deep ocean, where they may feed on large fishes. Remains of both giant squid species have been found in the stomachs of sperm whales, which

are probably their only natural predator. Scientists first photographed *A. dux* in the wild in 2005 while it was attacking baited hooks at a depth of 900m. *M. hamiltoni* has yet to be observed in nature. Overall, much remains to be learned about these marine giants.

Protecting Freshwater and Terrestrial Molluscs

Species extinction rates have increased dramatically in the last 400 years, raising concern that a sixth, human-caused mass extinction may be under way (see Concept 25.4). Among the many taxa under threat, molluscs have the dubious distinction of being the animal group with the largest number of documented extinctions **(Figure 33.21)**.

Figure 33.21

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The silent extinction.
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A pie chart is given, and next to that are a dozen workers on a mound of pearl mussels killed to make buttons, circa 1919.

Pie chart data are as follows:

Animal Group	Approximate proportion of pie chart
Molluscs	40 percent
Birds	21 percent
Fishes	18 percent
Mammals	14 percent
Insects	8 percent
Amphibians	6 percent
Reptiles (excluding birds)	3 percent
Other invertebrates	2 percent

Molluscs account for a largely unheralded but sobering 40% of all documented extinctions of animal species. These extinctions have resulted from human actions such as habitat loss and overharvesting. Many pearl mussel populations, for example, were driven to extinction by overharvesting for their shells, which were

used to make buttons and other goods. Land snails are highly vulnerable to the same threats; like pearl mussels, they are among the world's most imperiled animal groups.

MAKE CONNECTIONS *Freshwater bivalves feed on and can reduce the abundance of photosynthetic protists and bacteria. As such, what types of effects would the extinction of freshwater bivalves likely have on aquatic communities (see*Concept 28.6*)?**

*For suggested answer, see*Appendix A.**

Threats to molluscs are especially severe in two groups: freshwater bivalves and terrestrial gastropods. For example, the pearl mussels, a group of freshwater bivalves that can make natural pearls (gems that form when a mussel or oyster secretes layers of a lustrous coating around a grain of sand or other small irritant), are among the world's most endangered animals. Roughly 10% of the 300 pearl mussel species that once lived in North America have become extinct in the last 100 years, and over two-thirds of those that remain are threatened by extinction. Terrestrial gastropods, such as the snail in Figure 33.21, are faring no better. Hundreds of Pacific island land snails have disappeared since 1800. Overall, more than 50% of the Pacific island land snails are extinct or under imminent threat of extinction.

Threats faced by freshwater and terrestrial molluscs include habitat loss, pollution, competition or predation by non-native species, and overharvesting by humans. Is it too late to protect these molluscs? In some locations, reducing water pollution and changing how water is released from dams have led to dramatic rebounds in pearl mussel populations. Such results provide hope that with corrective measures, other endangered mollusc species can be revived.

Annelids

Annelida means "little rings," referring to the annelid body's resemblance to a series of fused rings. Annelids are segmented worms that live in the sea, in most freshwater habitats, and in damp soil. Annelids, which have a coelom (and no hemocoel), range in length from less than 1mm to more than 3m.

Traditionally, the phylum Annelida was divided into three main groups, Polychaeta (the polychaetes), Oligochaeta (the oligochaetes), and Hirudinea (the leeches). The names of the first two of these groups reflected the relative number of chaetae, bristles made of chitin, on their bodies: polychaetes (from the Greek *poly*, many, and *chaitē*, long hair) have many more chaetae per segment than do oligochaetes.

However, a 2011 phylogenomic study and other recent molecular analyses have indicated that the oligochaetes are a subgroup of the polychaetes, making the polychaetes (as defined morphologically) a paraphyletic group. Likewise, the leeches have been shown to be a subgroup of the oligochaetes. As a result, these traditional names are no longer used to describe the evolutionary history of the annelids. Instead,

current evidence indicates that the annelids can be divided into two major clades, Errantia and Sedentaria—a grouping that reflects broad differences in lifestyle.

Errantians

Clade Errantia (from the Old French *errant*, traveling) is a large and diverse group, most of whose members are marine **(Figure 33.22)**. As their name suggests, many errantians are mobile; some swim among the plankton (small, drifting organisms), while many others crawl on or burrow in the seafloor. Many are predators, while others are grazers that feed on large, multicellular algae. The group also includes some relatively immobile species, such as the tube-dwelling *Platynereis*, a marine species that recently has become a model organism for studying neurobiology and development.

Figure 33.22

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An errantian, the predator *Nereimyra punctata.*
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Photograph of Nereimyra punctata. Segmented, worm shaped body with parapodia on each segment (here, parapodia appear as long, fine, string-like structures).

This marine annelid ambushes prey from burrows it has constructed on the seafloor. N. punctata hunts by touch, detecting its prey with long sensory organs called cirri that extend from the burrow.

In many errantians, each body segment has a pair of prominent paddle-like or ridge-like structures called parapodia ("beside feet") that function in locomotion (see Figure 33.22). Each parapodium has numerous chaetae. (Possession of parapodia with numerous chaetae is not unique to Errantia, however, as some members of the other major clade of annelids, Sedentaria, also have these features.) In many species, the parapodia are richly supplied with blood vessels and also function as gills. Errantians also tend to have well-developed jaws and sensory organs, as might be expected of predators or grazers that move about in search of food.

Sedentarians

Species in the other major clade of annelids, Sedentaria (from the Latin *sedere*, sit), tend to be less mobile than those in Errantia. Some species burrow slowly through marine sediments or soil, while others live within tubes that protect and support their soft bodies. Tube-dwelling sedentarians often have elaborate gills or tentacles used for filter feeding **(Figure 33.23)**.

Figure 33.23

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The Christmas tree worm, *Spirobranchus giganteus.*
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Photograph of Christmas tree worm under water. Pictured are two central stalks, each with branchlike, feathery level structures that resemble bright red Christmas trees.

This sedentarian's two tree-shaped whorls are tentacles, which it uses in gas exchange and to collect food particles from the water. The tentacles emerge from a calcium carbonate tube secreted by the worm that protects and supports its soft body.

Although the Christmas tree worm shown in Figure 33.23 once was classified as a "polychaete," current evidence indicates it is a sedentarian. The clade Sedentaria also contains former "oligochaetes," including the two groups we turn to next, the leeches and the earthworms.

Leeches

Some leeches are parasites that suck blood by attaching temporarily to other animals, including humans **(Figure 33.24)**, but most are predators that feed on other invertebrates. Leeches range in length from 1 to 30cm. Most leeches inhabit fresh water, but there are also marine species and terrestrial leeches, which live in moist vegetation. Some parasitic species use bladelike jaws to slit the skin of their host. The host is usually oblivious to this attack because the leech secretes an anesthetic. After making the incision, the leech secretes a chemical, hirudin, which keeps the blood of the host from coagulating near the incision. The parasite then sucks as much blood as it can hold, often more than ten times its own weight. After this gorging, a leech can last for months without another meal.

Figure 33.24

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A leech.
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Photograph of a leech attached to a human thumb. Leech has dark, slick, segmented worm-like body and is only a little smaller than the thumb.

A nurse applied this medicinal leech (Hirudo medicinalis) to a patient's sore thumb to drain blood from a hematoma (an abnormal accumulation of blood around an internal injury).

Until the 20th century, leeches were frequently used for bloodletting. Today they are used to drain blood that accumulates in tissues following certain injuries or surgeries. In addition, forms of hirudin produced with recombinant DNA techniques can be used to dissolve unwanted blood clots that form during surgery or as a result of heart disease.

Earthworms

Earthworms eat their way through the soil, extracting nutrients as the soil passes through the alimentary canal. Undigested material, mixed with mucus secreted into the canal, is eliminated as fecal castings through the anus. Farmers value earthworms because the animals till and aerate the earth, and their castings improve the texture of the soil. (Charles Darwin estimated that one acre of farmland contains about 50,000 earthworms, producing 18tons of castings per year.)

A guided tour of the anatomy of an earthworm, which is representative of annelids, is shown in **Figure 33.25**. Earthworms are hermaphrodites, but they do cross-fertilize. Two earthworms mate by aligning themselves in opposite directions in such a way that they exchange sperm, and then they separate. Some earthworms can also reproduce asexually by fragmentation followed by regeneration.

Figure 33.25

Anatomy of an earthworm, a sedentarian.

Diagram of earthworm, with structures and functions labeled. Next to the diagram is a photograph of a man holding a giant Australian earthworm, which appears at least two feet long.

Each segment is surrounded by longitudinal muscle, which in turn is surrounded by circular muscle. Earthworms coordinate the contraction of these two sets of muscles to move. Parts are labeled as follows.

Coelom. The coelom of the earthworm is partitioned by septa.

Metanephridium. Each segment of the worm contains a pair of excretory tubules, called metanephridia, that discharge wastes from the blood and coelomic fluid through exterior pores. Tiny blood vessels are abundant in the earthworm's skin, which functions as its respiratory organ.

Ventral nerve cords. The nerve cords penetrate the septa and run the length of the animal, as do the digestive tract and longitudinal blood vessels.

Chaetae. Each segment has four pairs of chaetae, bristles that provide traction for burrowing. Many of the internal structures are repeated within each segment of the earthworm.

Cerebral ganglia. The earthworm nervous system features a brainlike pair of cerebral ganglia above and in front of the pharynx. A ring of nerves around the pharynx connects to a subpharyngeal ganglion, from which a fused pair of nerve cords runs posteriorly. The circulatory system, a network of vessels, is closed. The dorsal and ventral vessels are linked by segmental pairs of vessels, some of which are muscular and pump blood through the circulatory system.

Diagram shows earthworm from mouth end towards anus end, detailing structures in between. There is also a cutaway cross-section of a segment of the earthworm labeled. Mouth is connected to tube that runs length of earthworm (ending at anus). Behind mouth is subpharyngeal ganglion attached to the ventral nerve cords (which run parallel to the alimentary canal) and the cerebral ganglia. Behind the mouth on the alimentary canal are the pharynx and then the esophagus. Blood vessels loop around the alimentary canal. Posterior to the esophagus is the crop, which is a bulge in the alimentary canal, and then the gizzard, followed by the intestine. Metanephridium are drawn adjacent to the alimentary canal. On the outer surface of the worm are tiny hair-like structures called chaetae; the clitellum is also on diagrammed as a bulge on the outer surface of the worm, closer to the head than the anus. The cross-section of the worm segment shows the longitudinal and circular muscles beneath the epidermis (skin), and there is a cuticle on the outer surface of the worm. The septum is a partition between segments. The alimentary canal is in the center of the worm's body and the dorsal vessel runs parallel. Fused nerve cords, nephrostomes, and metanephridium are also labeled.

As a group, Lophotrochozoa encompasses a remarkable range of body plans, as illustrated by members of such phyla as Syndermata, Ectoprocta, Mollusca, and Annelida. Next we'll explore the diversity of Ecdysozoa, a dominant presence on Earth in terms of sheer number of species.

Concept 33.4: Ecdysozoans are the most species-rich animal group

An evolutionary tree highlights ecdysozoa.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia.

Although defined primarily by molecular evidence, the clade Ecdysozoa includes animals that shed a **cuticle** (a tough external coat) as they grow. In fact, the group derives its name from this process, which is called *ecdysis*, or **molting**. Ecdysozoa includes about eight animal phyla and contains more known species than all other animal, protist, fungus, and plant groups combined. Here we'll focus on the two largest ecdysozoan phyla, the nematodes and arthropods, which are among the most successful and abundant of all animal groups.

Nematodes

Among the most ubiquitous of animals, nematodes (phylum Nematoda), or roundworms, are found in most aquatic habitats, in the soil, in the moist tissues of plants, and in the body fluids and tissues of animals. The cylindrical bodies of nematodes range from less than 1mm to more than 1m long, often tapering to a fine tip at the posterior end and to a blunter tip at the anterior end **(Figure 33.26)**. A nematode's body is covered by a tough cuticle (a type of exoskeleton); as the worm grows, it periodically sheds its old cuticle and secretes a new, larger one. Nematodes have an alimentary canal, though they lack a circulatory system. Nutrients are transported throughout the body via fluid in the hemocoel. The body wall muscles are all longitudinal, and their contraction produces a thrashing motion.

Figure 33.26

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A free-living nematode.
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Nematode with a long, thin, whiplike body.

(colorized SEM)

Multitudes of nematodes live in moist soil and in decomposing organic matter on the bottoms of lakes and oceans. While 25,000 species are known, perhaps 20 times that number actually exist. These free-living worms play an important role in decomposition and nutrient cycling, but little is known about most species. One species of soil nematode, *Caenorhabditis elegans*, however, is very well studied and has become a model research organism in biology (see Concept 47.3). Ongoing studies of *C. elegans* are providing insight into mechanisms involved in aging in humans, as well as many other topics.

Phylum Nematoda includes many species that parasitize plants, and some are major agricultural pests that attack the roots of crops. Other nematodes parasitize animals. Some of these species benefit humans by attacking insects such as cutworms that feed on the roots of crop plants. On the other hand, humans are hosts to at least 50 nematode species, including various pinworms and hookworms. One notorious nematode is *Trichinella spiralis*, the worm that causes trichinosis **(Figure 33.27)**. Humans acquire this nematode by eating raw or undercooked pork or other meat (including wild game such as bear or walrus) that has juvenile worms encysted in the muscle tissue. Within the human intestines, the juveniles develop into sexually mature adults. Females burrow into the intestinal muscles and produce more juveniles, which bore through the body or travel in lymphatic vessels to other organs, including skeletal muscles, where they encyst.

Figure 33.27

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    Juveniles of the parasitic nematode *Trichinella spiralis*
encysted in human muscle tissue.
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Cysts (round structures) within human muscle tissue contain multiple worm-shaped parasitic nematodes. Scale: 50 micrometers.

(LM)

Parasitic nematodes have an extraordinary molecular toolkit that enables them to redirect some of the cellular functions of their hosts. Some species inject their plant hosts with molecules that induce the development of root cells, which then supply nutrients to the parasites. When *Trichinella* parasitizes animals, it regulates the expression of specific muscle cell genes encoding proteins that make the cell elastic enough to house the nematode. Additionally, the infected muscle cell releases signals that promote the growth of new blood vessels, which then supply the nematode with nutrients.

Arthropods

Zoologists estimate that there are about a billion billion (1018) arthropods living on Earth. More than 1 million arthropod species have been described, most of which are insects. In fact, two out of every three known species are arthropods, and members of the phylum Arthropoda can be found in nearly all habitats of the biosphere. By the criteria of species diversity, distribution, and sheer numbers, arthropods must be regarded as the most successful of all animal phyla.

Arthropod Origins

Biologists hypothesize that the diversity and success of **arthropods** are related to their body plan—their segmented body, hard exoskeleton, and jointed appendages. How did this body plan arise and what advantages did it provide?

The earliest fossils of arthropods are from the Cambrian explosion (535–525 millionyears ago), indicating that the arthropods are at least that old. The fossil record of the Cambrian explosion also contains many species of *lobopods*, a group from which arthropods may have evolved. Lobopods such as *Hallucigenia* (see **Figure 32.7**) had segmented bodies, but most of their body segments were identical to one another. Early arthropods, such as the trilobites, also showed little variation from segment to segment **(Figure 33.28)**. As arthropods continued to evolve, groups of segments tended to become functionally united into “body regions” specialized for tasks such as feeding, walking, or swimming. These evolutionary changes resulted

not only in great diversification but also in efficient body plans that permit the division of labor among different body regions.

Figure 33.28

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A trilobite fossil.
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A trilobite fossil has a body that is roughly tear-shaped with many obvious segments and spiny structures lining one end.

Trilobites were common denizens of the shallow seas throughout the Paleozoic era but disappeared with the great Permian extinctions about 250 millionyears ago. Paleontologists have described about 4,000 trilobite species.

What genetic changes led to the increasing complexity of the arthropod body plan? Arthropods today have two unusual *Hox* genes, both of which influence segmentation. To test whether these genes could have driven the evolution of increased body segment diversity in arthropods, researchers studied *Hox* genes in onychophorans (see Figure 33.2), close relatives of arthropods **(Figure 33.29)**. Their results indicate that the diversity of arthropod body plans did *not* arise from the acquisition of new *Hox* genes. Instead, the evolution of body segment diversity in arthropods was probably driven by changes in the sequence or regulation of existing *Hox* genes (see Concept 25.5).

Figure 33.29

Inquiry: Did the arthropod body plan result from new *Hox*genes?

Experiment

One hypothesis suggests that the arthropod body plan resulted from the origin (by gene duplication and subsequent mutations) of two unusual *Hox* genes found in arthropods: *Ultrabithorax* (*Ubx*) and *abdominal-A* (*abd-A*). Researchers tested this hypothesis using onychophorans, a group of invertebrates closely related to arthropods. Unlike many living arthropods, onychophorans have a body plan in which most body segments are identical to one another. If the origin of the *Ubx* and *abd-A Hox* genes drove the evolution of

body segment diversity in arthropods, these genes probably arose on the arthropod branch of the evolutionary tree:

A phylogenetic tree. A branch begins with a common ancestor of onychophorans and arthropods. The branch that goes to arthropods has a point labeled Origin of U b x and a b d, A Hox genes?.

According to this hypothesis, *Ubx* and *abd-A* would not have been present in the common ancestor of arthropods and onychophorans; hence, onychophorans should not have these genes. The researchers examined the *Hox* genes of the onychophoran *Acanthokara kaputensis*.

Results

The onychophoran *A. kaputensis* has all arthropod *Hox* genes, including *Ubx* and *abd-A*.

A micrograph of a onychophoran embryo. Antenna, jaws, and L 1 through 15 body segments are highlighted. Red in L 15 indicates the body regions of this onychophoran embryo in which U b x or a b d, A genes were expressed.

Conclusion

The evolution of increased body segment diversity in arthropods was not related to the origin of new *Hox* genes.

Data from J. K. Grenier et al., Evolution of the entire arthropod *Hox* gene set predated the origin and radiation of the onychophoran/arthropod clade, *Current Biology* 7:547–553 (1997).

WHAT IF? *Suppose*A. kaputensis*did not have the*Ubx*and*abd-A***Hox*genes. How would the conclusions of this study have been affected? Explain.*

*For suggested answer, see***Appendix A*.*

General Characteristics of Arthropods

Over the course of evolution, the appendages of some arthropods have become modified, specializing in functions such as walking, feeding, sensory reception, reproduction, and defense. Like the appendages from which they were derived, these modified structures are jointed and come in pairs. **Figure 33.30** illustrates the diverse appendages and other arthropod characteristics of a lobster.

Figure 33.30

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External anatomy of an arthropod.
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A diagram of a lobster.

Many of the distinctive features of arthropods are apparent in this dorsal view of a lobster. The body is segmented, but this character is obvious only in the post-genital region or "tail," located behind the genitals. The appendages (including antennae, pincers, mouthparts, walking legs, and swimming appendages) are jointed. The head bears a pair of compound (multilens) eyes. The whole body, including appendages, is covered by an exoskeleton.

Major body segments include the cephalothorax (head with eyes and thorax) and the abdomen. Four walking legs are visible from the side view, along with swimming appendages (one pair per body segment), pincers (for defense), and antennae (for sensory reception).

The body of an arthropod is completely covered by the cuticle, an exoskeleton constructed from layers of protein and the polysaccharide chitin. As you know if you've ever eaten a crab or lobster, the cuticle can be thick and hard over some parts of the body and thin and flexible over others, such as the joints. The rigid exoskeleton protects the animal and provides points of attachment for the muscles that move the appendages. But it also prevents the arthropod from growing, unless it occasionally sheds its exoskeleton and produces a larger one. This molting process is energetically expensive, and it leaves the arthropod vulnerable to predation and other dangers until its new, soft exoskeleton hardens.

When the arthropod exoskeleton first evolved in the sea, its main functions were probably protection and anchorage for muscles, but it later enabled certain arthropods to live on land. The exoskeleton's relative impermeability to water helped prevent desiccation, and its strength provided support when arthropods left the buoyancy of water. Fossil evidence suggests that arthropods were among the first animals to colonize land, roughly 450 millionyears ago. These fossils include fragments of arthropod remains, as well as possible millipede burrows. Arthropod fossils from several continents indicate that by 410 millionyears ago, millipedes, centipedes, spiders, and a variety of wingless insects all had colonized land.

Arthropods have well-developed sensory organs, including eyes, olfactory (smell) receptors, and antennae that function in both touch and smell. Most sensory organs are concentrated at the anterior end of the animal, although there are interesting exceptions. Female butterflies, for example, "taste" plants using sensory organs on their feet.

Like many molluscs, arthropods have an **open circulatory system**, in which fluid called *hemolymph* is propelled by a heart through short arteries and then into the hemocoel—the body cavity surrounding the tissues and organs. (The term *blood* is generally reserved for fluid in a closed circulatory system.) Hemolymph reenters the arthropod heart through pores that are usually equipped with valves. In most

arthropods, the coelom that forms in the embryo becomes much reduced as development progresses, and the hemocoel becomes the main body cavity in adults.

A variety of specialized gas exchange organs have evolved in arthropods. These organs allow the diffusion of respiratory gases in spite of the exoskeleton. Most aquatic species have gills with thin, feathery extensions that expose a large surface area to the surrounding water. Terrestrial arthropods generally have internal surfaces specialized for gas exchange. For example, most insects have tracheal systems, branched air ducts leading into the interior of the body from pores in the cuticle.

Morphological and molecular data suggest that living arthropods consist of three major lineages that diverged early in the phylum's evolution: **chelicerates** (sea spiders, horseshoe crabs, scorpions, ticks, mites, and spiders); **myriapods** (centipedes and millipedes); and **pancrustaceans** (a recently defined, diverse group that includes insects as well as lobsters, shrimp, barnacles, and other crustaceans).

Chelicerates

Chelicerates (clade Chelicerata) are named for clawlike feeding appendages called **chelicerae**, which serve as pincers or fangs. Chelicerates lack antennae, and most have simple eyes (eyes with a single lens).

The earliest chelicerates were **eurypterids**, or water scorpions. These marine and freshwater predators grew up to 3m long; it is thought that some species could have walked on land, much as land crabs do today. Most of the marine chelicerates, including all of the eurypterids, are extinct. Among the marine chelicerates that survive today are the sea spiders (pycnogonids) and horseshoe crabs **(Figure 33.31)**.

Figure 33.31

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    Horseshoe crabs (*Limulus polyphemus*).
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Horseshoe crabs on a beach.

Common on the Atlantic and Gulf coasts of the United States, these "living fossils" have changed little in hundreds of millions of years. They are surviving members of a rich diversity of chelicerates that once filled the seas.

The bulk of modern chelicerates are **arachnids**, a group that includes scorpions, spiders, ticks, and mites **(Figure 33.32)**. Nearly all ticks are bloodsucking parasites that live on the body surfaces of reptiles or mammals. Parasitic mites live on or in a wide variety of vertebrates, invertebrates, and plants.

Figure 33.32

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Arachnids.
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Scorpion has coiled tail and extended pincers. Dust mite has dome-like body (scale: 50 micrometers). Spider with bright coloration is on a web.

Scorpions have pedipalps that are pincers specialized for defense and the capture of food. The tip of the tail bears a poisonous stinger. Dust mites are ubiquitous scavengers in human dwellings but are harmless except to those people who are allergic to them (colorized SEM). Web-building spiders are generally most active during the daytime.

Arachnids have six pairs of appendages: the chelicerae; a pair of appendages called *pedipalps* that function in sensing, feeding, defense, or reproduction; and four pairs of walking legs. Spiders use their fang-like chelicerae, which are equipped with poison glands, to attack prey. As the chelicerae pierce the prey, the spider secretes digestive juices onto the prey's torn tissues. The food softens, and the spider sucks up the liquid meal. In most spiders, gas exchange is carried out by **book lungs**, stacked platelike structures contained in an internal chamber **(Figure 33.33)**. The extensive surface area of these respiratory organs enhances the exchange of O2 and CO2 between the hemolymph and air.

Figure 33.33

```
Book lungs.
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Spider body plan. Pedipalps, chelicera, heart and book lungs are highlighted.

A unique adaptation of many spiders is the ability to catch insects by constructing webs of silk, a liquid protein produced by specialized abdominal glands. The silk is spun by organs called spinnerets into fibers that then solidify. Each spider engineers a web characteristic of its species and builds it perfectly on the first try, indicating that this complex behavior is inherited. Various spiders also use silk in other ways: as droplines for rapid escape, as a cover for eggs, and even as "gift wrap" for food that males offer females during courtship. Many small spiders also extrude silk into the air and let themselves be transported by wind, a behavior known as "ballooning."

Myriapods

Millipedes and centipedes belong to the clade Myriapoda **(Figure 33.34)**. All living myriapods are terrestrial. The myriapod head has a pair of antennae and three pairs of appendages modified as mouthparts, including the jaw-like mandibles.

Figure 33.34

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Myriapods.
```

Part a. Millipede: thin black segmented body with legs extending from each segment. Part b. Centipede: brownish segmented body, shorter and with fewer legs than the millipede.

Millipedes have a large number of legs, though fewer than the thousand their name implies. Each trunk segment is formed from two fused segments and bears two pairs of legs (see Figure 33.34a). Millipedes eat decaying leaves and other plant matter. They may have been among the earliest animals on land, living on mosses and early vascular plants.

Unlike millipedes, centipedes are carnivores. Each segment of a centipede's trunk region has one pair of legs (see Figure 33.34b). Centipedes have poison claws on their foremost trunk segment that paralyze prey and aid in defense.

Pancrustaceans

A series of recent papers present evidence that terrestrial insects are more closely related to lobsters and other crustaceans than they are to the terrestrial group we just discussed, the myriapods (millipedes and centipedes). These studies also suggest that the diverse group of organisms referred to as crustaceans are paraphyletic: Some lineages of crustaceans are more closely related to insects than they are to other crustaceans **(Figure 33.35)**. However, together the insects and crustaceans form a clade, which systematists have named Pancrustacea (from the Greek *pan*, all). We turn next to a description of the members of Pancrustacea, focusing first on crustaceans and then on the insects.

Figure 33.35

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The phylogenetic position of the insects.
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Tree diagram for the phylogenetic position of the insects.

Recent results have shown that the insects are nested within lineages of aquatic crustaceans. The remipedians are one of several groups of aquatic crustaceans that may be the sister group to the insects.

Tree diagram represented as a list.

Common ancestor of arthropods (branch point 1).

Branch point 2.

Branch point 3.

Branch point 4. Insects (example: beetle). (dotted line to) Remipedians (a crustacean group). Other crustaceans (example: lobster). Myriapods (example: centipede). Chelicerates (example: scorpion)

DRAW IT *Circle the portions of this tree that comprise the clade Pancrustacea.*

*For suggested answer, see*Appendix A.**

Crustaceans

Crustaceans (crabs, lobsters, shrimps, barnacles, and many others) thrive in a broad range of marine, freshwater, and terrestrial environments. Many crustaceans have highly specialized appendages. Lobsters and crayfishes, for instance, have a toolkit of 19 pairs of appendages (see **Figure 33.30**). The anterior-most appendages form two pairs of antennae; crustaceans are the only arthropods with two pairs. Three or more pairs of appendages are modified as mouthparts, including the hard mandibles. Walking legs are present on the thorax, and, unlike their terrestrial relatives, the insects, crustaceans also have appendages on their post-genital region, or "tail."

Small crustaceans exchange gases across thin areas of the cuticle; larger species have gills. Nitrogenous wastes also diffuse through thin areas of the cuticle, but a pair of glands regulates the salt balance of the hemolymph.

Sexes are separate in most crustaceans. In the case of lobsters and crayfishes, the male uses a specialized pair of abdominal appendages to transfer sperm to the reproductive pore of the female during copulation. Most aquatic crustaceans go through one or more swimming larval stages.

One of the largest groups of crustaceans (numbering over 11,000 species) is the *isopods*, which include terrestrial, freshwater, and marine species. Some isopod species are abundant in habitats at the bottom of the deep ocean. Among the terrestrial isopods are the pill bugs, or wood lice, common on the undersides of moist logs and leaves.

Lobsters, crayfishes, crabs, and shrimps are all relatively large crustaceans called *decapods***(Figure 33.36)**. The cuticle of decapods is hardened by calcium carbonate. Most decapod species are marine. Crayfishes, however, live in fresh water, and some tropical crabs live on land.

Figure 33.36

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A ghost crab, an example of a decapod.
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Ghost crab is a pale brownish-white color.

Ghost crabs live on sandy ocean beaches worldwide. Primarily nocturnal, they take shelter in burrows during the day.

Many small crustaceans are important members of marine and freshwater plankton communities. Planktonic crustaceans include many species of *copepods*, which are among the most numerous of all animals. Some copepods are grazers that feed upon algae, while others are predators that eat small animals (including smaller copepods!). Copepods are rivaled in abundance by the shrimplike krill, which grow to about 5cm long **(Figure 33.37)**. A major food source for baleen whales (including blue whales, humpbacks, and right whales), krill are now being harvested in great numbers by humans for food and agricultural fertilizer. The larvae of many larger-bodied crustaceans are also planktonic.

Figure 33.37

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Krill.
```

A krill swims.

These planktonic crustaceans are consumed in vast quantities by some whales.

With the exception of a few parasitic species, barnacles are a group of sessile crustaceans whose cuticle is hardened into a shell containing calcium carbonate **(Figure 33.38)**. Most barnacles anchor themselves to rocks, boat hulls, pilings, and other submerged surfaces. Their natural adhesive is as strong as synthetic glues. These barnacles feed by extending appendages from their shell to strain food from the water. Barnacles were not recognized as crustaceans until the 1800s, when naturalists discovered that barnacle larvae resemble the larvae of other crustaceans. The remarkable mix of unique traits and crustacean homologies found in barnacles was a major inspiration to Charles Darwin as he developed his theory of evolution.

Figure 33.38

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Barnacles.
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Small appendages extend from a barnacle.

The jointed appendages projecting from the barnacles' shells capture organisms and organic particles suspended in the water.

We turn now to a group nested within the paraphyletic crustaceans, the insects.

Insects

Insects and their six-legged terrestrial relatives form an enormous clade, Hexapoda; we'll focus here on the insects, since as a group they have more described species than all other eukaryotic groups combined. Insects live in almost every terrestrial habitat and in fresh water, and flying insects fill the air. Insects are rare, though not absent, in marine habitats. The internal anatomy of an insect includes several complex organ systems, which are highlighted in **Figure 33.39**.

Figure 33.39

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Anatomy of a grasshopper, an insect.
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Grasshopper diagram illustrates insect body plan.

The insect body has three regions: head, thorax, and post-genital region. The segmentation of the thorax and post-genital region is obvious, but the segments that form the head are fused.

Parts are labeled as follows.

Cerebral ganglia. The two nerve cords meet in the head, where the ganglia of several anterior segments are fused into a brain (colored white below). The antennae, eyes, and other sense organs are concentrated on the head. Insect mouthparts are formed from several pairs of modified appendages. The mouthparts include mandibles, which grasshoppers use for chewing. In other insects, mouthparts are specialized for lapping, piercing, or sucking.

Nerve cords. The insect nervous system consists of a pair of ventral nerve cords with several segmental ganglia.

Tracheal tubes. Gas exchange in insects is accomplished by a tracheal system of branched, chitin-lined tubes that infiltrate the body and carry oxygen directly to cells. The tracheal system opens to the outside of the body through spiracles, pores that can control air flow and water loss by opening or closing.

Heart. The insect heart drives hemolymph through an open circulatory system.

Malpighian tubules. Metabolic wastes are removed from the hemolymph by excretory organs called Malpighian tubules, which are outpocketings of the digestive tract.

A grasshopper diagram illustrates insect body plan. Head with compound eyes and antennae is anterior to thorax, which is anterior to abdomen. Internally, beginning in the head, the cerebral ganglion, insect

mouthparts, crop, ovary, vagina and anus are also labeled. Nerve cords run the length of the body, as do vessels (dorsal artery is labeled, which connects to the heart). Malpighian tubules branch in the posterior region. Tracheal tubes labeled near center of body.

The oldest insect fossils date to about 415 millionyears ago. Later, an explosion in insect diversity took place when insect flight evolved during the Carboniferous and Permian periods (359–252 millionyears ago). An animal that can fly can escape predators, find food and mates, and disperse to new habitats more effectively than an animal that must crawl about on the ground. Many insects have one or two pairs of wings that emerge from the dorsal side of the thorax. Because the wings are extensions of the cuticle, insects can fly without sacrificing any walking legs **(Figure 33.40)**. By contrast, the flying vertebrates—birds and bats—have one of their two pairs of walking legs modified into wings, making some of these species clumsy on the ground.

Figure 33.40

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Ladybird beetle in flight.
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Photograph shows both wings and some legs extended while ladybird beetle in flight.

Insects also radiated in response to the origin of new plant species, which provided new sources of food. By the speciation mechanisms described in Concept 24.2, an insect population feeding on a new plant species can diverge from other populations, eventually forming a new species of insect. A fossil record of diverse insect mouthparts, for example, suggests that specialized modes of feeding on gymnosperms and other Carboniferous plants contributed to early adaptive radiations of insects. Later, a major increase in insect diversity appears to have been stimulated by the evolutionary expansion of flowering plants during the mid-Cretaceous period (about 100 millionyears ago). Although insect and plant diversity decreased during the Cretaceous mass extinction, both groups have rebounded over the past 66 millionyears. Increases in the diversity of particular insect groups have often been associated with radiations of the flowering plants on which they fed.

Many insects undergo metamorphosis during their development. In the **incomplete metamorphosis** of grasshoppers and some other insect groups, the young (called nymphs) resemble adults but are smaller, have different body proportions, and lack wings. The nymph undergoes a series of molts, each time looking more like an adult. With the final molt, the insect reaches full size, acquires wings, and becomes sexually mature. Insects with **complete metamorphosis** have larval stages specialized for eating and growing that are known by such names as caterpillar, maggot, or grub. The larval stage looks entirely different from the adult stage, which is specialized for dispersal and reproduction. Metamorphosis from the larval stage to the adult occurs during a pupal stage **(Figure 33.41)**.

Figure 33.41

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Complete metamorphosis of a butterfly.
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The stages of becoming a butterfly are a larva (caterpillar), b pupa, c later-stage pupa, (d) emerging adult, and (e) adult.

(a) *The larva (caterpillar) spends its time eating and growing, molting as it grows.* ***(b)*** *After several molts, the larva develops into a pupa.* ***(c)*** *Within the pupa, the larval tissues are broken down, and the adult is built by the division and differentiation of cells that were quiescent in the larva.* ***(d)*** *Eventually, the adult begins to emerge from the pupal cuticle.* ***(e)*** *Hemolymph is pumped into veins of the wings and then withdrawn, leaving the hardened veins as struts supporting the wings. The insect will fly off and reproduce, deriving much of its nourishment from the food reserves stored by the feeding larva.*

Video

Butterfly Emerging

image

Play

image

Play

Reproduction in insects is usually sexual, with separate male and female individuals. Adults come together and recognize each other as members of the same species by advertising with bright colors (as in butterflies), sounds (as in crickets), or odors (as in moths). Fertilization is generally internal. In most species, sperm are deposited directly into the female's vagina at the time of copulation, though in some species the male deposits a sperm packet outside the female, and the female picks it up. An internal structure in the female called the spermatheca stores the sperm, usually enough to fertilize more than one batch of eggs. Many insects mate only once in a lifetime. After mating, a female often lays her eggs on an appropriate food source where the next generation can begin eating as soon as it hatches.

Insects are classified in more than 30 orders, 8 of which are introduced in **Figure 33.42**.

Figure 33.42

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Exploring Insect Diversity
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Exploring insect diversity.

Although there are more than 30 orders of insects, we'll focus on just 8 here. Two orders of wingless insects, the bristletails (Archaeognatha) and silverfish (Zygentoma), diverged from other insects early in insect evolution. Evolutionary relationships among the other groups discussed here are under debate and so are not depicted on the tree. Tree diagram represented as a list.

Branch point 1. Archaeognatha (bristletails; 350 species). Example is a photograph of a wingless insect on wood.

Branch point 2. Zygentoma (silverfish; 450 species). Example is a wingless insect with long antennae. Winged insects (many orders).

Complete Metamorphosis. Winged insects in this category include the Coleoptera (male snout weevil with very long mouthpart shown); the Diptera (red tachinid shown, a fly-like insect); the Hymenoptera (wasp pictured); and the Lepidoptera (hawkmoth that strongly resembles a hummingbird with long proboscis pictured).

Coleoptera (beetles; 350,000 species): Beetles, such as this male snout weevil (Rhiastus lasternus), constitute the most species-rich order of insects. They have two pairs of wings, one of which is thick and stiff, the other membranous. They have an armored exoskeleton and mouthparts adapted for biting and chewing.

Diptera (151,000 species): Dipterans have one pair of wings; the second pair has become modified into balancing organs called halteres. Their mouthparts are adapted for sucking, piercing, or lapping. Flies and mosquitoes are among the best-known dipterans, which live as scavengers, predators, and parasites. Like many other insects, flies such as this red tachinid (Adejeania vexatrix) have well-developed compound eyes that provide a wide-angle view and excel at detecting fast movements.

Hymenoptera (125,000 species): Most hymenopterans, which include ants, bees, and wasps, are highly social insects. They have two pairs of membranous wings, a mobile head, and chewing or sucking mouthparts. The females of many species have a posterior stinging organ. Many species, such as this European paper wasp (Polistes dominulus), build elaborate nests.

Lepidoptera (120,000 species: Butterflies and moths have two pairs of wings covered with tiny scales. To feed, they uncoil a long proboscis, visible in this photograph of a hummingbird hawkmoth (Macroglossum stellatarum). This moth's name refers to its ability to hover in the air while feeding from a flower. Most lepidopterans feed on nectar, but some species feed on other substances, including animal blood or tears.

Incomplete metamorphosis. Winged insects in this category include the Hemiptera (adult stink bug guarding a cluster of small offspring pictured); and the Orthoptera (spear-bearer katydid pictured;

appendages are covered in spiny points and the face has three striking white dots and a very large mouthpart and smaller protrusion from the top of the head, both of which resemble flowers).

Hemiptera (85,000 species): Hemipterans include so-called true bugs, such as stink bugs, bed bugs, and assassin bugs. (Insects in other orders are sometimes erroneously called bugs.)

Hemipterans have two pairs of wings, one pair partly leathery, the other pair membranous. They have piercing or sucking mouthparts and undergo incomplete metamorphosis, as shown in this image of an adult stink bug guarding its offspring (nymphs).

Orthoptera (13,000 species): Grasshoppers, crickets, and their relatives are mostly herbivorous. They have large hind legs adapted for jumping, two pairs of wings (one leathery, one membranous), and biting or chewing mouthparts. This aptly named spiny devil katydid (Panacanthus cuspidatus) has a face and legs specialized for making a threatening display. Male orthopterans commonly make courtship sounds by rubbing together body parts, such as ridges on their hind legs.

Animals as numerous, diverse, and widespread as insects are bound to affect the lives of most other terrestrial organisms, including humans. Insects consume enormous quantities of plant matter; play key roles as predators, parasites, and decomposers; and are an essential source of food for larger animals such as lizards, rodents, and birds. Humans depend on bees, flies, and many other insects to pollinate crops and orchards. In addition, people in many parts of the world eat insects as an important source of protein. On the other hand, insects are carriers for many diseases, including African sleeping sickness (spread by tsetse flies that carry the protist *Trypanosoma*; see **Figure 28.9**) and malaria (spread by mosquitoes that carry the protist *Plasmodium*; see **Figure 23.18** and **Figure 28.18**).

Insects also compete with humans for food. In parts of Africa, for instance, insects claim about 75% of the crops. In the United States, billions of dollars are spent each year on pesticides. But insect populations often evolve pesticide resistance after short periods of time. Try as we may, not even humans have challenged the preeminence of insects and their arthropod kin. As one prominent entomologist put it: “Bugs are not going to inherit the Earth. They own it now. So we might as well make peace with the landlord.”

Concept 33.5: Echinoderms and chordates are deuterostomes

An evolutionary tree highlights deuterostomia.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia.

Sea stars, sea urchins, and other echinoderms (phylum Echinodermata) may seem to have little in common with vertebrates (animals that have a backbone) and other members of phylum Chordata. Nevertheless, DNA evidence indicates that echinoderms and chordates are closely related, with both phyla belonging to the Deuterostomia clade of bilaterian animals. Echinoderms and chordates also share features characteristic of a deuterostome mode of development, such as radial cleavage and formation of the anus from the blastopore (see Figure 32.10). As discussed in Concept 32.4, however, some animal phyla with members that have deuterostome developmental features, including ectoprocts and brachiopods, are not in the deuterostome clade. Hence, despite its name, the clade Deuterostomia is defined primarily by DNA similarities, not developmental similarities.

Echinoderms

Sea stars (commonly called starfish) and most other groups of **echinoderms** (from the Greek *echin*, spiny, and *derma*, skin) are slow-moving or sessile marine animals. Echinoderms have a coelom. A thin epidermis covers an endoskeleton of hard calcareous plates, and most species are prickly from skeletal bumps and spines. Unique to echinoderms is the **water vascular system**, a network of hydraulic canals branching into extensions called **tube feet** that function in locomotion and feeding **(Figure 33.43)**. Sexual reproduction of echinoderms usually involves separate male and female individuals that release their gametes into the water.

Figure 33.43

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Anatomy of a sea star, an echinoderm (top view).
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Anatomy of sea star. Anus and stomach are in central disk., ringed by ring canal. Gonads are clustered in one arm. Radial nerve extends to tip of each tube foot.

In the diagram, the digestive glands, radial canal, and gonads are illustrated in separate arms for clarity, but each is present in every arm. The photograph shows a sea star surrounded by sea urchins, which are also echinoderms.

Anatomy of a sea star.

Central disk. The central disk has a nerve ring and nerve cords radiating from the ring into the arms.

Digestive glands secrete digestive juices and aid in the absorption and storage of nutrients. A short digestive tract runs from the mouth on the bottom of the central disk to the anus on top of the disk. The surface of a sea star is covered by spines that help defend against predators, as well as by small gills that provide gas exchange.

Madreporite. Water can flow in or out of the water vascular system into the surrounding water through the madreporite.

Radial canal. The water vascular system consists of a ring canal in the central disk and five radial canals, each running in a groove down the entire length of an arm. Branching from each radial canal are hundreds of hollow, muscular tube feet filled with fluid. Each tube foot consists of a bulb-like ampulla and a podium (foot portion). When the ampulla squeezes, water is forced into the podium, which expands and contacts the substrate. Adhesive chemicals are then secreted from the base of the podium, attaching it to the substrate. To detach the tube foot, de-adhesive chemicals are secreted and muscles in the podium contract, forcing water back into the ampulla and shortening the podium. As it moves, a sea star leaves an observable footprint of adhesive material on the substrate.

Video

Echinoderm Tube Feet

image

Play

image

Play

Echinoderms descended from bilaterally symmetrical ancestors, yet on first inspection most species seem to have a radially symmetrical form. The internal and external parts of most adult echinoderms radiate from the center, often as five spokes. However, echinoderm larvae have bilateral symmetry. Furthermore, the symmetry of adult echinoderms is not truly radial. For example, the opening (madreporite) of a sea star's water vascular system is not central but shifted to one side.

Living echinoderms are divided into five clades.

Asteroidea: Sea Stars and Sea Daisies

Sea stars have arms radiating from a central disk; the undersurfaces of the arms bear tube feet. By a combination of muscular and chemical actions, the tube feet can attach to or detach from a substrate. The sea star adheres firmly to rocks or creeps along slowly as its tube feet extend, grip, release, extend, and grip again. Although the base of the tube foot has a flattened disk that resembles a suction cup, the gripping action results from adhesive chemicals, not suction (see Figure 33.43).

Sea stars also use their tube feet to grasp prey, such as clams and oysters. The arms of the sea star embrace the closed bivalve, clinging tightly with their tube feet. The sea star then turns part of its stomach inside out, everting it through its mouth and into the narrow opening between the halves of the bivalve's shell. Next, the digestive system of the sea star secretes juices that begin digesting the mollusc within its own shell. The sea star then brings its stomach back inside its body, where digestion of the mollusc's (now liquefied) body is completed. The ability to begin the digestive process outside of its body allows a sea star to consume bivalves and other prey species that are much larger than its mouth.

Sea stars and some other echinoderms have considerable powers of regeneration. Sea stars can regrow lost arms, and members of one genus can even regrow an entire body from a single arm if part of the central disk remains attached.

The clade Asteroidea, to which sea stars belong, also includes a small group of armless species, the *sea daisies*. Only three species of sea daisies are known, all of which live on submerged wood. A sea daisy's body is typically disk-shaped; it has a five-sided organization and measures less than a centimeter in diameter **(Figure 33.44)**. The edge of the body is ringed with small spines. Sea daisies absorb nutrients through a membrane that surrounds their body.

Figure 33.44

A sea daisy (clade Asteroidea).

A sea daisy is a flat, circular creature with symmetry.

Ophiuroidea: Brittle Stars

Brittle stars have a distinct central disk and long, flexible arms **(Figure 33.45)**. They move primarily by lashing their arms in serpentine movements. The base of a brittle star tube foot lacks the flattened disk found in sea stars but does secrete adhesive chemicals. Hence, like sea stars and other echinoderms, brittle stars can use their tube feet to grip substrates. Some species are suspension feeders; others are predators or scavengers.

Figure 33.45

```
A brittle star (clade Ophiuroidea).
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A brittle star is a small, star-shaped creature with 5 noodle-like appendages.

Echinoidea: Sea Urchins and Sand Dollars

Sea urchins and sand dollars have no arms, but they do have five radially arranged groups of tube feet that function in slow movement. Sea urchins also have muscles that pivot their long spines, which aid in locomotion as well as protection **(Figure 33.46)**. A sea urchin's mouth, located on its underside, is ringed by highly complex, jaw-like structures that are well adapted to eating seaweed. Sea urchins are roughly spherical, whereas sand dollars are flat disks.

Figure 33.46

```
A sea urchin (clade Echinoidea).
```

A sea urchin is a ball-like creature with a star pattern.

Crinoidea: Sea Lilies and Feather Stars

Sea lilies live attached to the substrate by a stalk; feather stars crawl about by using their long, flexible arms. Both use their arms in suspension feeding. The arms encircle the mouth, which is directed upward, away from the substrate **(Figure 33.47)**. Crinoidea is an ancient group whose morphology has changed little over the course of evolution; fossilized sea lilies some 500 millionyears old are extremely similar to present-day members of the clade.

Figure 33.47

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A feather star (clade Crinoidea).
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A feather star is an elaborate create with fiddlehead-like appendages.

Holothuroidea: Sea Cucumbers

On casual inspection, sea cucumbers do not look much like other echinoderms. They lack spines, and their endoskeleton is much reduced. They are also elongated in their oral-aboral axis, giving them the shape for which they are named and further disguising their relationship to sea stars and sea urchins **(Figure 33.48)**.

Closer examination, however, reveals that sea cucumbers have five radially arranged sections of tube feet, as in other echinoderms. Some of the tube feet around the mouth are developed as feeding tentacles.

Figure 33.48

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A sea cucumber (clade Holothuroidea).
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A sea cucumber is a slug-like creature.

Chordates

Phylum Chordata consists of two basal groups of invertebrates, the lancelets and the tunicates, as well as the vertebrates. Chordates are bilaterally symmetrical animals with a coelom and segmented bodies. The close relationship between echinoderms and chordates does not mean that one phylum evolved from the other. In fact, echinoderms and chordates have evolved independently of one another for over 500 millionyears. We will trace the phylogeny of chordates in Chapter 34, focusing on the history of vertebrates.